Momentum Regime Shifts: A Bayesian Hidden Markov Model Approach to Cross-Sectional Stock Selection

By

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A thesis submitted in partial fulfillment of the requirements for the degree of Master of Science in

Quantitative Finance and Actuarial Science

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Date: May 2026

Abstract

This thesis investigates how a real-time regime signal interacts with cross-sectional momentum and finds that momentum is not a single stable anomaly but a regime-dependent family of horizon-specific effects: in calm markets, intermediate-horizon continuation dominates the cross-section; in panic, reversal takes over as beaten-down stocks across all horizons become more likely to outperform. A Bayesian Hidden Markov Model produces a filtered probability of market panic, which is combined with momentum at 12 lookback horizons in machine learning models to rank stocks. The regime signal functions as a context variable – it carries no direct linear return-predictive content but conditions how the model jointly uses the full momentum term structure in a given market state. Out-of-sample on 2011–2025 U.S. equities, only a nonlinear model (XGBoost) exploits this structure (Sharpe 1.11, six-factor alpha 24.7%, $t = 4.78$); both a classification-based and a regression-based linear model collapse, establishing that the interaction is inherently nonlinear. Cross-sectional z-score analysis confirms this regime-dependent structure in the raw data. The regime signal is essential (removing it reduces the Sharpe by more than 60%) and most valuable during stress, with the long-short spread widening from 1.3% to 3.1% per month. These findings are consistent with the interpretation that momentum's well-documented fragility arises partly from applying unconditional models to what is fundamentally a regime-dependent phenomenon.

Acknowledgements

I would like to thank my supervisor, Denis Kojevnikov, for his guidance and feedback throughout this project. I am also grateful to Tilburg University's School of Economics and Management for providing access to WRDS data. Any remaining errors are my own.

Management Summary

Cross-sectional momentum – buying recent winners and selling recent losers – is one of the most profitable and most fragile anomalies in finance. This thesis finds that momentum's fragility may arise because it is not a single stable pattern but a regime-dependent family of effects: in calm markets, stocks that rose over intermediate horizons continue to outperform; in panic, stocks that have fallen across all lookback horizons become more likely to outperform. The regime signal, derived from a Bayesian Hidden Markov Model estimated on real-time stress indicators, functions as a context variable that conditions how the model jointly uses the full momentum term structure rather than predicting returns directly. Only a nonlinear model (XGBoost) can exploit this structure; linear models with identical features fail entirely, including both a classification-based and a regression-based specification.

The thesis develops a two-stage framework (Section 3). In the first stage, a Bayesian two-state Hidden Markov Model is estimated via Gibbs sampling on four macrofinancial stress indicators (drawdown, dispersion, relative market participation, and credit spread), producing a filtered probability of market panic. In the second stage, this regime signal is combined with momentum at 12 lookback horizons in models of increasing complexity: a deterministic formula, logistic regression, Ridge regression, and XGBoost.

Out-of-sample on 167 months of U.S. equity data (2011–2025), only XGBoost survives long-short construction (Sharpe 1.11, six-factor alpha 24.7%). The regime signal is essential: removing it reduces the Sharpe by more than 60%. Cross-sectional z-score analysis confirms that momentum's predictive structure shifts with market conditions in the raw data, independent of any model. Stress tests show the strategy can absorb a 12-month recession but is vulnerable to prolonged fundamental deterioration where beaten-down stocks never recover.

Limitations include the single test period (2011–2025), US-only sample, and untested resilience to depression-style downturns. The thesis complements the exposure-scaling approach of Daniel and Moskowitz (2016) with a stock-selection channel: rather than adjusting how much to hold, the regime signal adjusts which stocks to hold.

Contents

1 Introduction

Cross-sectional momentum, the tendency for recent stock market winners to continue outperforming and losers to continue underperforming, is one of the most robust anomalies in empirical finance (Jegadeesh and Titman, 1993). Yet it is also one of the most fragile: momentum portfolios suffered losses exceeding 50% during the 2009 market rebound (Daniel and Moskowitz, 2016). This tension between long-run profitability and vulnerability to regime shifts motivates the central question of this thesis: how does a real-time regime signal interact with the cross-sectional momentum anomaly, and what does this interaction reveal about momentum's underlying structure?

Existing responses to momentum's fragility adjust either exposure or horizon weights based on backward-looking indicators (Section 2). This thesis proposes an alternative: a Bayesian Hidden Markov Model estimated on real-time stress indicators produces a filtered probability of market panic, which is fed alongside 12 momentum horizons into machine learning models that learn how to adapt cross-sectional stock selection to the current regime.

The central claim is that this regime signal is not a direct return predictor but a *context variable*: it conditions how the model jointly uses the full momentum term structure in a given market state. No prior study has documented how the full term structure of cross-sectional momentum predictability reorganises across regimes, or that exploiting this reorganisation requires nonlinear modelling.

Out-of-sample evaluation on 167 months of U.S. equity data (2011–2025) yields three main findings. First, only XGBoost produces meaningful long-short returns (Sharpe 1.11, six-factor alpha 24.7%, $t = 4.78$); logistic regression with identical features collapses (Sharpe -0.03), establishing that the interaction is inherently nonlinear. Second, the regime signal is essential – removing it reduces the Sharpe by more than 60% – and most valuable during stress, with the long-short spread widening from 1.3% to 3.1% per month in panic. Third, cross-sectional z-score analysis confirms this shift: in calm, the model's long leg selects stocks with a humped momentum profile peaking at intermediate horizons; in panic, the same leg selects stocks uniformly below average at every horizon from 1 to 12 months.

Specifically, this thesis addresses three questions: (i) does the regime signal's interaction with momentum require nonlinear modelling, and if so, why does a linear model fail? (ii) how does the cross-sectional structure of momentum change between calm and panic regimes? (iii) how robust is the regime-momentum interaction to alternative signal specifications, risk preferences, and prolonged market stress?

The thesis makes three contributions. First, it documents how the full term structure of cross-sectional momentum predictability reorganises between calm and panic regimes, with short-horizon signals flipping sign during stress. Second, it proposes and validates a framework in which a macrofinancial regime signal functions as a context variable for stock selection, conditioning how the model jointly weighs the full momentum term structure rather than predicting returns directly. Third, it establishes that this interaction is inherently nonlinear: both classification-based and regression-based linear models fail on identical features, while a tree-based model succeeds.

The remainder of the thesis is organised as follows. Section 2 reviews the momentum anomaly, regime-switching models, market-state momentum, and machine learning in asset pricing. Section 3 develops the two-stage framework: the Bayesian HMM for regime classification and the three cross-sectional stock selection methods. Section 4 describes the data, HMM feature selection procedure, and cross-sectional features. Section 5 presents the out-of-sample results, including regime-conditional performance, stress tests, SHAP analysis, ablation studies, and the risk-aversion analysis. Section 6 summarises the findings, discusses limitations, and suggests directions for future research.

2 Literature Review

The following review surveys four strands of literature that this thesis builds upon: the momentum anomaly and its empirical properties, momentum crashes and risk management, regime-switching models in finance, and machine learning approaches to cross-sectional asset pricing.

2.1 The Momentum Anomaly

Jegadeesh and Titman (1993) established cross-sectional momentum as a pervasive feature of equity markets: portfolios that buy past winners and sell past losers earn significant returns over 3-to-12-month holding periods. Their original strategy, which ranks stocks by their cumulative returns over the past 12 months (skipping the most recent month to avoid microstructure effects) and holds the resulting long-short portfolio for 3 to 12 months, has since been replicated across international markets, asset classes, and sample periods (Jegadeesh and Titman, 2001; Asness et al., 2013).

The economic mechanisms behind momentum remain debated. Behavioural explanations emphasise investor underreaction to news, where information diffuses slowly into prices due to cognitive biases such as conservatism and anchoring (Barberis et al., 1998), or due to gradual information diffusion across investor networks (Hong and Stein, 1999), generating predictable continuation in returns. Risk-based explanations argue that momentum profits compensate for exposure to systematic risks that vary over time, though no single risk factor has been shown to fully explain the anomaly (Fama and French, 2015).

A key empirical feature of momentum is its sensitivity to the lookback horizon. While the standard 12-month lookback captures intermediate-term continuation, shorter lookbacks (1–3 months) tend to capture short-term reversal effects (Jegadeesh and Titman, 1993), and lookbacks beyond 12 months reflect long-term mean reversion (De Bondt and Thaler, 1985). This horizon dependence suggests that momentum is not a single signal but a family of return patterns whose relative strength varies with market conditions, a hypothesis that this thesis tests directly.

2.2 Momentum Crashes and Risk Management

Despite its long-run profitability, momentum is prone to spectacular crashes. Daniel and Moskowitz (2016) document that momentum portfolios experienced catastrophic losses during the 1932 and 2009 market rebounds. They show that these crashes are predictable: they occur when the market recovers sharply after a period of poor performance, and the conditional mean and volatility of momentum returns are strongly related to past market states. Specifically, when the market has recently declined, past losers (which the momentum strategy shorts) tend to be high-beta stocks that rally disproportionately during recoveries, while past winners (which the strategy holds long) are low-beta defensive stocks that lag.

Barroso and Santa-Clara (2015) propose a risk-management approach: scaling momentum exposure inversely with the strategy's recent realised volatility. This constant-volatility overlay substantially improves the Sharpe ratio by reducing exposure precisely when crash risk is elevated. Their key insight is that momentum's conditional volatility is highly predictable and strongly time-varying, so a simple volatility-targeting rule captures much of the available improvement.

The liquidity literature provides a theoretical basis for why momentum crashes reverse: margin spirals (Brunnermeier and Pedersen, 2009) and institutional flow pressure (Vayanos and Woolley, 2013) create temporary price dislocations during stress that revert when selling pressure subsides, a mechanism central to this thesis's interpretation of the regime-momentum interaction.

These contributions frame the regime-switching approach pursued in this thesis. Rather than scaling exposure based on the strategy's own volatility (a backward-looking measure), this thesis uses a real-time regime signal, derived from contemporaneous stress indicators, to adapt the composition of the momentum portfolio, changing which stocks are selected rather than how much capital is deployed.

2.3 Regime-Switching Models in Finance

Hidden Markov Models have a long history in financial econometrics, originating with Hamilton (1989), who introduced the Markov-switching model for business cycle analysis. Albert and Chib (1993) subsequently developed Bayesian estimation of these models via

Gibbs sampling, demonstrating that MCMC methods can efficiently handle the joint estimation of latent states and model parameters. Their approach, which this thesis adapts to a multivariate setting with four stress indicators, remains the standard framework for Bayesian regime-switching analysis.

In the context of equity markets, HMMs have been used to identify bull and bear market regimes (Rydén et al., 1998), model time-varying volatility, and detect structural breaks in return distributions (Guidolin and Timmermann, 2006). A common finding is that equity returns are well described by a two-state model with a high-mean, low-volatility state (bull market) and a low-mean, high-volatility state (bear market), where the bear state is less persistent but more volatile (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007). This asymmetry in regime characteristics has important implications for portfolio management, as the transition from calm to panic states is associated with a simultaneous increase in downside risk and a breakdown of traditional diversification (Guidolin and Timmermann, 2007).

In most prior work, the estimated regime is used to make binary portfolio decisions – for example, reducing equity exposure during bear states (Ang and Bekaert, 2002) or switching between asset classes (Guidolin and Timmermann, 2007). The regime model in this thesis differs in two ways. First, rather than fitting the HMM to equity returns directly, it uses four macrofinancial stress indicators as observable inputs, allowing the regime classification to reflect multiple dimensions of market distress simultaneously. Second, the regime signal is not used to make binary decisions but is passed as a continuous feature to downstream machine learning models that learn its optimal use for cross-sectional stock selection.

2.4 Market-State Momentum: Goulding et al. (2023)

The most directly related work is Goulding et al. (2023), who condition cross-sectional momentum on the market state. They classify each month into one of four market cycles (Bull, Correction, Bear, Rebound) based on the trailing 1-month and rolling 12-month market returns. In each cycle, a different blending parameter a determines the weight given to 1-month versus 12-month momentum for ranking stocks: the composite signal is $a \cdot mom_1 + (1 - a) \cdot mom_{12}$, where $a = 0$ corresponds to pure 12-month momentum and $a = 1$ to pure 1-month momentum. Their dynamic (DYN) strategy selects the optimal blending parameters for each non-Bull state via grid search on the training sample.

This thesis extends Goulding et al. (2023) in three directions. First, the regime classification uses a probabilistic model (the HMM) with real-time stress indicators rather than a deterministic rule based on trailing market returns. The HMM produces a continuous panic probability rather than a discrete four-state classification, allowing for smoother portfolio transitions. Second, the cross-sectional models use the full momentum term structure (12 lookback horizons) rather than blending only two momentum signals. Third, the use of machine learning models (logistic regression and XGBoost) allows the data to determine how the regime signal interacts with the other features, rather than imposing a parametric blending rule.

The empirical analysis (Section 5) tests whether these methodological extensions translate into improved stock selection. Since Goulding et al. (2023) design their strategies for long-only construction, a direct performance comparison in long-short is not appropriate. Instead, the GHM market-cycle variable is used as an alternative regime signal within the XGBoost framework (Table 6), providing a fair test of whether trailing-return-based regime definitions capture the same information as the HMM's stress indicators.

2.5 Machine Learning in Cross-Sectional Asset Pricing

The use of machine learning methods for cross-sectional stock selection has grown rapidly in recent years. Gu et al. (2020) provide a comprehensive comparison of machine learning methods for predicting cross-sectional stock returns, finding that tree-based models and neural networks substantially outperform linear models by capturing nonlinearities and interactions among firm characteristics. Gradient-boosted tree models, including XGBoost (Chen and Guestrin, 2016), have proven particularly effective for tabular financial data because they handle missing values natively, capture nonlinear relationships and feature interactions without explicit specification, and are robust to irrelevant features. In cross-sectional return prediction, tree-based models consistently outperform linear models when the true data-generating process involves interactions between firm characteristics and macroeconomic conditions (Gu et al., 2020).

A central challenge in applying machine learning to asset pricing is interpretability. SHAP values (Lundberg and Lee, 2017; Lundberg et al., 2020), based on cooperative game theory, provide a principled decomposition of each prediction into feature-level contributions. For tree-based models, the TreeSHAP algorithm computes exact Shapley values in polynomial time, enabling detailed analysis of which features drive predictions and how their importance varies across observations. This thesis uses SHAP values to interpret how the regime signal enters XGBoost's predictions, an analysis that would be infeasible without model-agnostic interpretability tools.

While Cooper et al. (2004) show that momentum profits depend on market state and Novy-Marx (2012) demonstrates that the intermediate-horizon component drives momentum profits, no study has examined how the *full term structure* of cross-sectional momentum predictability reorganises across regimes. The framework developed in this thesis addresses this gap.

3 Methodology

The two-stage empirical framework underlying the analysis is developed as follows. The first stage estimates market regimes using a Bayesian Hidden Markov Model (Section 3.1). A two-state HMM is fit to four macrofinancial stress indicators via Gibbs sampling, producing a monthly filtered probability π_t^{filter} that quantifies how likely the market is to be in a panic state. Because this probability conditions only on information available at time t, it is a real-time signal with no look-ahead bias.

The second stage uses this regime signal to guide cross-sectional stock selection (Section 3.2). Three portfolio construction methods of increasing complexity serve distinct analytical roles. The deterministic formula (Method 0) tests whether simply varying the momentum lookback horizon with market conditions adds value. The logistic regression (Method 1) tests whether a linear model can exploit the regime signal alongside momentum. XGBoost (Method 2) tests whether nonlinear interactions between the regime signal and other features matter. Comparing M1 and M2 is the key diagnostic: if the regime signal functions as a context variable rather than a direct return predictor, a linear model should be unable to use it, while a tree-based model should find it important.

3.1 Stage 1: Regime Classification via Hidden Markov Model

3.1.1 Model Specification

This thesis employs a two-state ($K = 2$) Bayesian Hidden Markov Model to classify the market into latent regimes. The choice of $K = 2$ is validated empirically: Appendix E.4 shows that increasing to $K = 3, 4,$ or 5 states does not improve downstream portfolio performance and introduces estimation instability. The hidden state $s_t \in \{0, 1\}$ represents the unobserved regime at month t, where $s_t = 0$ denotes the *calm* regime and $s_t = 1$ denotes the *panic* regime. Crucially, as emphasized by Hamilton (1989), market regimes are not directly observable: an investor cannot know with certainty whether the market is currently in a calm or panic state. While regime classifications may appear obvious in retrospect, distinguishing regimes in real time is inherently difficult. The HMM formalises this uncertainty by treating regimes as hidden states that must be inferred probabilistically from observable market data (see Figure 1).

Observation equation Conditional on the regime, the vector of standardized features $\mathbf{z}_t \in \mathbb{R}^D$ (where $D = 4$) is drawn from a regime-specific multivariate Gaussian. The shorthand $\mathbf{z}_{1:t} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_t)$ denotes the full history of observations up to and including month t:

$$(\mathbf{z}_t \mid s_t = 0) \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \qquad (\mathbf{z}_t \mid s_t = 1) \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1). \tag{1}$$

Each regime is characterized by its own mean vector $\boldsymbol{\mu}_k \in \mathbb{R}^D$ and covariance matrix

$\mathbf{\Sigma}_k \in \mathbb{R}^{D\times D}$, where $k \in \{0 \text{ (calm)}, 1 \text{ (panic)}\}$. The multivariate specification allows the model to capture not only level differences in individual features across regimes, but also changes in the correlation structure; for instance, stress features may become more correlated during panic periods. The Gaussian assumption is standard in the HMM literature. Appendix F.7 verifies its adequacy by re-estimating the model with multivariate Student-t emissions: regime classifications agree on at least 97.8% of months, filtered panic probabilities correlate at ≥ 0.98, and BIC does not improve, confirming that the Gaussian specification is robust for this application.

Transition equation The evolution of the hidden state follows a first-order Markov chain with a 2×2 transition matrix:

$$\mathbf{P} = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix}, \tag{2}$$

where $P_{ij} = P(s_t = j \mid s_{t-1} = i)$ and each row sums to one. The diagonal elements P_{00} (calm persistence) and P_{11} (panic persistence) govern how likely each regime is to continue into the next month. The expected duration of the calm regime is $1/(1 - P_{00})$ months, and likewise $1/(1 - P_{11})$ for the panic regime.

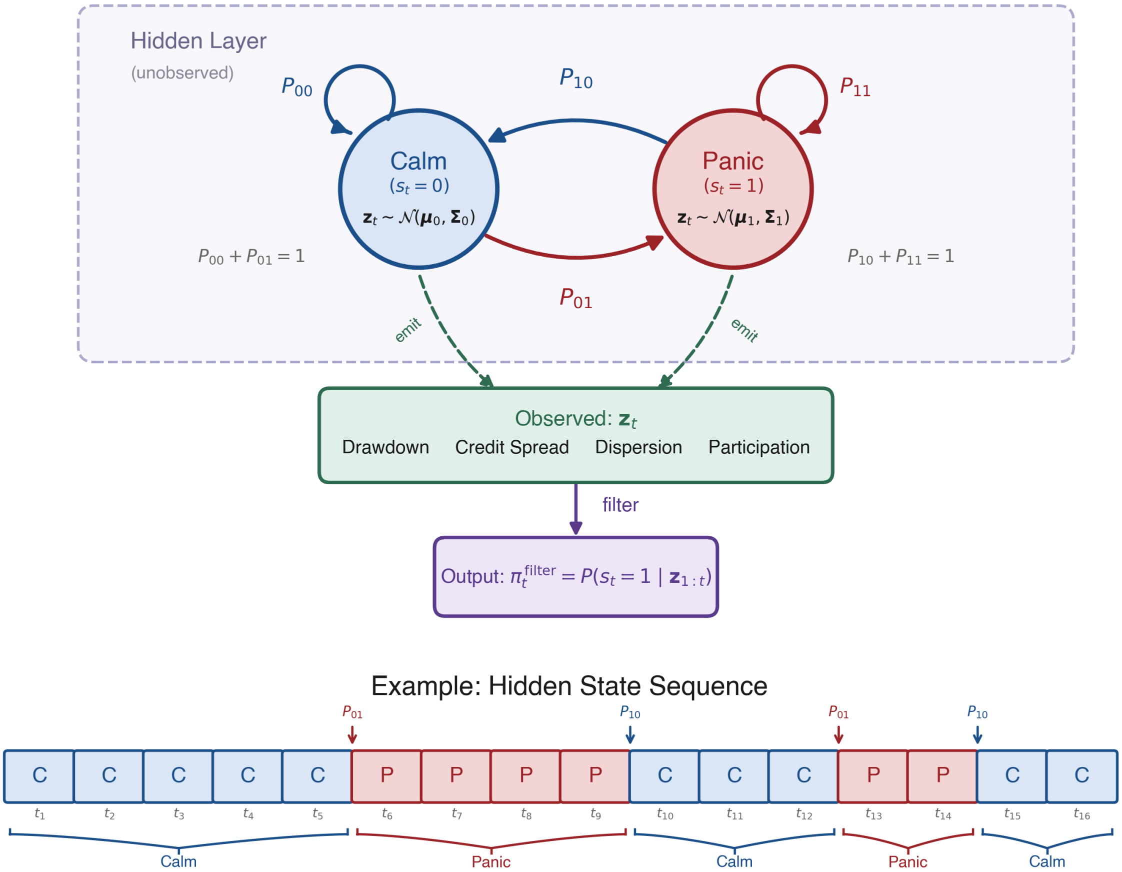


Figure 1: Architecture of the two-state Hidden Markov Model. Hidden states (Calm/Panic) evolve via transition probabilities P_{ij} and emit the four-dimensional observation vector $\mathbf{z}_t$. The forward filter inverts this process to produce $\pi_t^{\text{filter}} = P(s_t = 1 \mid \mathbf{z}_{1:t})$. The lower panel shows an example state sequence illustrating regime persistence and abrupt transitions.

3.1.2 Bayesian Estimation via Gibbs Sampling

The standard frequentist alternative is the Baum–Welch algorithm (Baum et al., 1970), a special case of Expectation-Maximisation (Dempster et al., 1977), which finds maximum likelihood point estimates. The Bayesian approach instead draws from the joint posterior:

$$p\left(\{s_t\}_{t=1}^T,\ \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=0}^1,\ \mathbf{P} \,\middle|\, \mathbf{z}_{1:T}\right).$$

This posterior is not available in closed form because the hidden states and model parameters depend on each other. Therefore we employ Gibbs sampling (Albert and Chib, 1993; Hoff, 2009), a Markov chain Monte Carlo (MCMC) method, which resolves this by alternating between three conditional updates: (i) the full hidden state path $\{s_t\}_{t=1}^T$, (ii) the emission parameters $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ for each regime, and (iii) the transition matrix

P. Conjugate priors ensure that each block can be sampled exactly from its conditional posterior.

3.1.3 Prior Specification

The priors encode two economic beliefs. First, before seeing the data, neither regime should be forced to have extreme values in any feature; the model should learn the characteristics of calm and panic states from the data. Second, regimes should be persistent rather than switching every few months.

Emission parameters For each regime k, the mean vector and covariance matrix are given a conjugate Normal-Inverse-Wishart (NIW) prior:[1]

$$\boldsymbol{\Sigma}_k \sim \mathcal{IW}(\nu_0, \boldsymbol{\Psi}_0), \qquad \boldsymbol{\mu}_k \mid \boldsymbol{\Sigma}_k \sim \mathcal{N}\left(\mathbf{m}_0, \frac{\boldsymbol{\Sigma}_k}{\kappa_0}\right). \tag{3}$$

The hyperparameters are chosen as

$$\mathbf{m}_0 = \mathbf{0}, \qquad \kappa_0 = 0.01, \qquad \nu_0 = D + 2, \qquad \boldsymbol{\Psi}_0 = \mathbf{I}_D(\nu_0 - D - 1), \tag{4}$$

where $D = 4$ is the number of features. After observing data, these prior hyperparameters update to posterior counterparts κ_n, $\mathbf{m}_n$, ν_n, and $\boldsymbol{\Psi}_n$ (Equation 9). Each hyperparameter controls a different aspect of the prior belief:

- $\mathbf{m}_0$ is the prior guess for each regime's mean vector. Setting $\mathbf{m}_0 = \mathbf{0}$ reflects that, since all features are standardized, there is no reason to expect either regime to be centered away from zero before seeing the data.

- κ_0 controls how strongly the prior mean $\mathbf{m}_0$ is trusted relative to the data. The very small value $\kappa_0 = 0.01$ makes this prior nearly uninformative, so the posterior mean is driven overwhelmingly by the observed data.

- ν_0 is the degrees of freedom of the Inverse-Wishart prior, interpretable as the number of pseudo-observations the prior is worth. The value $\nu_0 = D + 2 = 6$ is the smallest integer for which the prior expected value exists, yielding the most diffuse valid prior; a larger choice (e.g., $\nu_0 = 9$) would impose stronger confidence that the covariance is close to $\mathbf{I}_D$. With roughly 150 calm and 100 panic months in training, the posterior degrees of freedom ($\nu_n = \nu_0 + n_k$) far exceed the prior, so the data dominate the covariance estimates.

[1] The Inverse-Wishart is a probability distribution over positive-definite matrices, making it the natural prior for a covariance matrix: every sample is guaranteed to be symmetric and positive-definite. The Normal-Inverse-Wishart combines this with a conditional Normal prior on the mean. The prior is called *conjugate* because when the data are Gaussian, the posterior takes the same NIW form with updated hyperparameters, making each Gibbs update an exact draw rather than an approximation.

- $\mathbf{\Psi}_0$ is the scale matrix of the Inverse-Wishart prior. It is calibrated so that the prior expected covariance equals the identity matrix, $\mathbb{E}[\mathbf{\Sigma}_k] = \mathbf{\Psi}_0/(\nu_0 - D - 1) = \mathbf{I}_D$, implying unit marginal variance and no cross-feature correlation before observing the data.

Transition matrix Each row of the transition matrix is assigned an independent Dirichlet prior:

$$\mathbf{P}_{0,\cdot} \sim \mathrm{Dir}(9, 1), \qquad \mathbf{P}_{1,\cdot} \sim \mathrm{Dir}(1, 9). \tag{5}$$

Equivalently, in matrix form,

$$\boldsymbol{\alpha}_{\mathrm{dir}} = \begin{pmatrix} 9 & 1 \\ 1 & 9 \end{pmatrix}.$$

This prior favors persistence in both regimes: absent data, the prior mean transition matrix is

$$\mathbb{E}[\mathbf{P}] = \begin{pmatrix} 0.90 & 0.10 \\ 0.10 & 0.90 \end{pmatrix}. \tag{6}$$

In other words, the prior expects each regime to continue with probability 0.9, corresponding to an expected duration of roughly 10 months. This prior is informative enough to discourage pathological solutions but weak enough to be overridden by the data.

3.1.4 Gibbs Sampling Algorithm

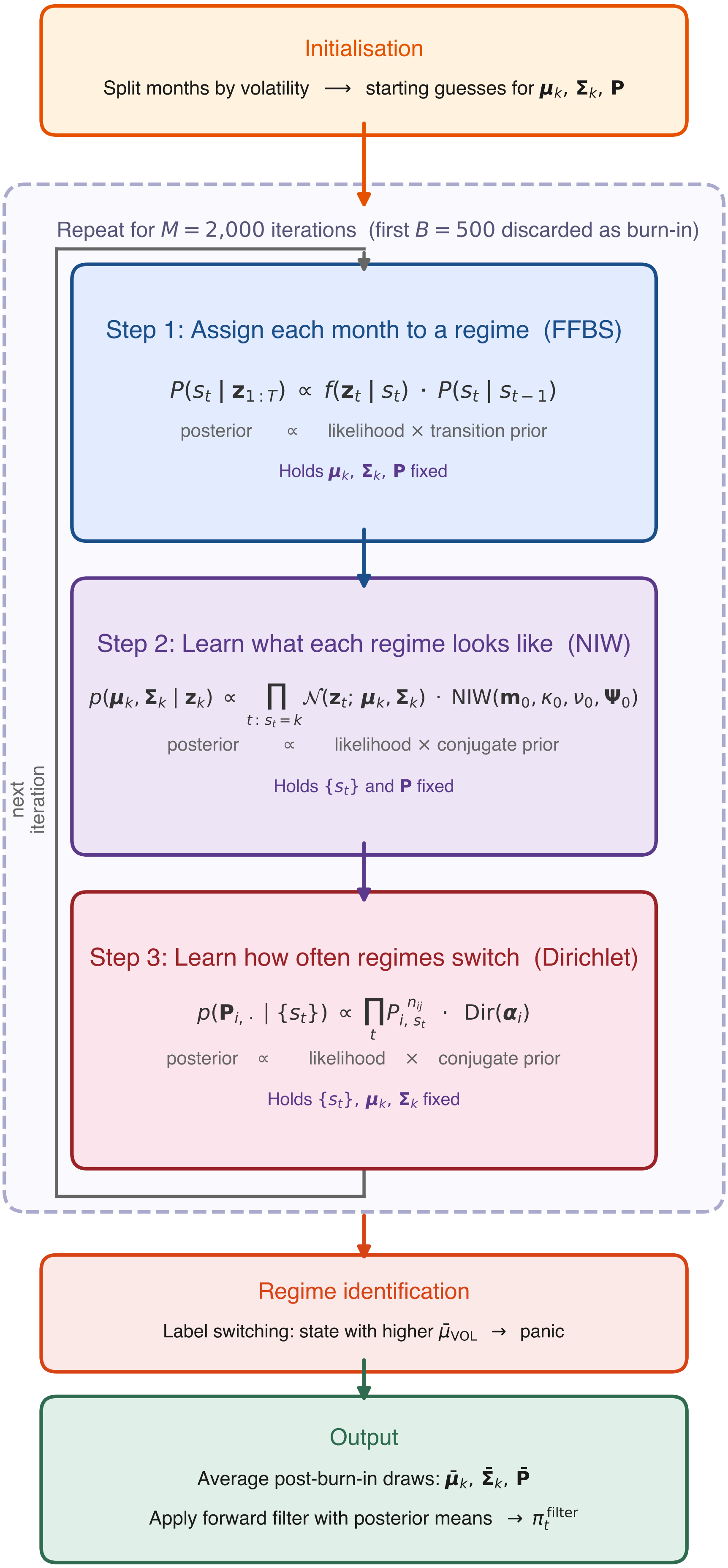
Initialization The Gibbs sampler requires starting values for the emission parameters and the transition matrix. A crude drawdown-based partition is used:[2] months with below-median drawdown DD_t^z (i.e., deeper drawdowns) are initially assigned to the panic state ($s_t = 1$), and the remaining months to the calm state ($s_t = 0$). The initial emission parameters are then computed from each group:

$$\boldsymbol{\mu}_k^{(0)} = \frac{1}{n_k} \sum_{t:\, s_t = k} \mathbf{z}_t, \qquad \mathbf{\Sigma}_k^{(0)} = \mathrm{Cov}(\{\mathbf{z}_t : s_t = k\}),$$

where n_k is the number of months assigned to regime k. The initial transition matrix is set to the prior mean, $\mathbf{P}^{(0)} = \mathbb{E}[\mathbf{P}]$ (Equation 6). These initial parameters give the first forward pass (described below) emission densities that can distinguish the two states.

Iterative updates Each iteration cycles through the three conditional updates described in Section 3.1.2. Figure 2 summarizes the cycle graphically.

[2] The choice of initialization has negligible impact on the posterior: the sampler is run for 2,000 iterations with 500 discarded as burn-in, and convergence diagnostics (Section 3.1.5) confirm that the chain mixes well regardless of the starting partition.



Initialisation
Split months by volatility ⟶ starting guesses for $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$
Repeat for $M = 2{,}000$ iterations (first $B = 500$ discarded as burn-in)
Step 1: Assign each month to a regime (FFBS)
$P(s_t \mid \mathbf{z}_{1:T}) \propto f(\mathbf{z}_t \mid s_t) \cdot P(s_t \mid s_{t-1})$
posterior ∝ likelihood × transition prior
Holds $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$ fixed
Step 2: Learn what each regime looks like (NIW)
$p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k \mid \mathbf{z}_k) \propto \prod_{t:\, s_t = k} \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \cdot \mathrm{NIW}(\mathbf{m}_0, \kappa_0, \nu_0, \boldsymbol{\Psi}_0)$
posterior ∝ likelihood × conjugate prior
Holds $\{s_t\}$ and $\mathbf{P}$ fixed
Step 3: Learn how often regimes switch (Dirichlet)
$p(\mathbf{P}_{i,\cdot} \mid \{s_t\}) \propto \prod_t P_{i,\, s_t}^{n_{ij}} \cdot \mathrm{Dir}(\boldsymbol{\alpha}_i)$
posterior ∝ likelihood × conjugate prior
Holds $\{s_t\}$, $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ fixed
next iteration
Regime identification
Label switching: state with higher $\bar{\mu}_{\mathrm{VOL}}$ → panic
Output
Average post-burn-in draws: $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, $\bar{\mathbf{P}}$
Apply forward filter with posterior means → π_t^{filter}

Step 1: Sampling the hidden state path via FFBS The full latent regime sequence $(s_1, \ldots, s_T)$ is sampled using Forward-Filtering Backward-Sampling (FFBS; see Frühwirth-Schnatter, 2006, ch. 11), holding the current emission parameters and transition matrix fixed.

The forward pass computes the filtered probability $\alpha_t(k) = P(s_t = k \mid \mathbf{z}_{1:t})$, the probability of being in state k at time t given all observations so far. This quantity is built in two sub-steps. First, a *prediction step* asks: before seeing this month's data, how likely is each state? Since the previous state is unknown, the algorithm sums over both possibilities, weighting each by its filtered probability from the previous month and the transition probability:

$$\alpha_{t|t-1}(k) = \alpha_{t-1}(0)\, P_{0k} + \alpha_{t-1}(1)\, P_{1k}.$$

Second, an *update step* revises this prediction once the current observation $\mathbf{z}_t$ arrives, using Bayes' rule: if the observed features look more like a panic month than a calm month, the probability of panic increases. Combining the two steps yields the recursion

$$\alpha_t(k) = \frac{f(\mathbf{z}_t \mid s_t = k)\, \alpha_{t|t-1}(k)}{f(\mathbf{z}_t \mid s_t = 0)\, \alpha_{t|t-1}(0) + f(\mathbf{z}_t \mid s_t = 1)\, \alpha_{t|t-1}(1)}, \tag{7}$$

where $f(\mathbf{z}_t \mid s_t = k) = \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the Gaussian emission density.[3] The recursion is initialized with $\alpha_0(k) = 1/2$, reflecting no prior preference for either regime.

The forward pass tells us how likely each regime is at every month, but if we simply flipped a weighted coin independently at each t we could produce implausible paths – for example, rapid calm-panic-calm-panic switching even though the transition matrix says regimes persist for roughly ten months. The backward pass prevents this by sampling states in reverse: once a future state has been drawn, the current state is sampled conditional on it, so the entire path respects the Markov transition dynamics.

The backward pass proceeds in two steps:

Terminal state At $t = T$, the filtered probability already conditions on all observations, so the terminal state is drawn directly:

$$s_T \sim \mathrm{Cat}(\alpha_T(0),\, \alpha_T(1)),$$

[3]In practice, these densities can be extremely small (or large) for four-dimensional observations, leading to numerical underflow. The implementation avoids this by working in log-space: log-likelihoods $\log f(\mathbf{z}_t \mid s_t = k)$ are computed directly, and the log-sum-exp identity $\log(e^a + e^b) = \max(a, b) + \log(1 + e^{-|a-b|})$ is used whenever probabilities must be normalised.

that is, the algorithm draws $s_T = 0$ (calm) with probability $\alpha_T(0)$ and $s_T = 1$ (panic) with probability $\alpha_T(1)$.[4]

Backward recursion For each earlier time step $t = T-1, T-2, \ldots, 1$, the algorithm chooses a state consistent with two pieces of information: (i) the filtered belief $\alpha_t(k)$ from the forward pass, and (ii) the already-sampled future state s_{t+1}, which constrains the current state via the transition probability $P_{k,s_{t+1}}$. Combining these two terms and normalising gives:

$$P(s_t = k \mid s_{t+1}, \mathbf{z}_{1:t}) = \frac{\alpha_t(k)\, P_{k,s_{t+1}}}{\alpha_t(0)\, P_{0,s_{t+1}} + \alpha_t(1)\, P_{1,s_{t+1}}}. \tag{8}$$

For example, if the forward filter assigns high probability to calm at time t ($\alpha_t(0)$ is large) and the already-sampled s_{t+1} is also calm (so $P_{0,0}$ is large), then the numerator for $k = 0$ dominates and the algorithm will very likely sample $s_t = 0$.

The output of the full FFBS procedure is one draw from the posterior over state paths. Across Gibbs iterations, different paths are sampled, exploring the full posterior uncertainty.

Importantly, the backward pass is used only during training to improve posterior parameter estimates. The trading signal π_t^{filter} relies solely on the causal forward filter.

Step 2: Updating the emission parameters Given the sampled state path, each regime k has n_k observations assigned to it. From these, two summary statistics are computed: the sample mean $\bar{\mathbf{x}}_k = \frac{1}{n_k}\sum_{t:\, s_t=k} \mathbf{z}_t$ and the scatter matrix $\mathbf{S}_k = \sum_{t:\, s_t=k}(\mathbf{z}_t - \bar{\mathbf{x}}_k)(\mathbf{z}_t - \bar{\mathbf{x}}_k)^\top$. The posterior hyperparameters are:

$$\kappa_n = \kappa_0 + n_k, \qquad \mathbf{m}_n = \frac{\kappa_0 \mathbf{m}_0 + n_k \bar{\mathbf{x}}_k}{\kappa_n}, \qquad \nu_n = \nu_0 + n_k,$$
$$\mathbf{\Psi}_n = \mathbf{\Psi}_0 + \mathbf{S}_k + \frac{\kappa_0 n_k}{\kappa_n}(\bar{\mathbf{x}}_k - \mathbf{m}_0)(\bar{\mathbf{x}}_k - \mathbf{m}_0)^\top. \tag{9}$$

Since $\kappa_0 = 0.01 \ll n_k$, the posterior is driven almost entirely by the data.

New regime parameters are then drawn from the updated posterior: $\mathbf{\Sigma}_k \sim \mathcal{IW}(\nu_n, \mathbf{\Psi}_n)$ and $\boldsymbol{\mu}_k \mid \mathbf{\Sigma}_k \sim \mathcal{N}(\mathbf{m}_n, \mathbf{\Sigma}_k/\kappa_n)$.

Step 3: Updating the transition matrix The sampled path is scanned to count transitions: $n_{ij} = \sum_{t=2}^{T} \mathbb{1}(s_{t-1} = i,\, s_t = j)$ records how many times the path moved from state i to state j. For example, if the sampled path contains 140 calm-to-calm transitions and 10 calm-to-panic transitions, then $n_{00} = 140$ and $n_{01} = 10$. These observed counts

[4] Cat denotes the categorical distribution, the standard notation for a single draw from a discrete probability vector. With two states it reduces to a weighted coin flip.

are added to the Dirichlet prior pseudo-counts to obtain the posterior:

$$\mathbf{P}_{i,\cdot} \mid \{s_t\} \sim \mathrm{Dir}\big(\alpha_{i0} + n_{i0},\ \alpha_{i1} + n_{i1}\big). \tag{10}$$

For row 0 (calm), the prior pseudo-counts are $(\alpha_{00}, \alpha_{01}) = (9, 1)$ from Equation (5), so the posterior becomes $\mathrm{Dir}(9 + 140,\ 1 + 10) = \mathrm{Dir}(149,\ 11)$. The expected persistence probability under this posterior is $149/160 = 0.93$, reflecting that the data reinforced the prior belief in regime persistence. A new transition matrix row is then drawn from this Dirichlet distribution, and the same procedure is applied independently to row 1 (panic).

With all three blocks updated, the sampler returns to Step 1. After sufficient iterations the draws converge to samples from the joint posterior.

3.1.5 Implementation and Convergence

The Gibbs sampler is estimated on the training sample only. To reduce sensitivity to MCMC initialization, the full sampler is run independently with $N_s = 200$ different random seeds. Each chain is run for $M = 2{,}000$ iterations with the initialization described above. The first $B = 500$ iterations are discarded as burn-in, leaving 1,500 posterior draws per seed.

For each seed, posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ are computed by averaging over the post-burn-in draws. A causal forward filter is then applied to the full sample using these parameters to obtain a seed-specific filtered probability $\pi_t^{\mathrm{filter},(s)}$. The final trading signal is the average across seeds:

$$\pi_t^{\mathrm{filter}} = \frac{1}{N_s} \sum_{s=1}^{N_s} \pi_t^{\mathrm{filter},(s)}. \tag{11}$$

This Bayesian model averaging over 200 seeds substantially reduces the variance of the regime signal. Cross-seed stability tests confirm that the resulting Sharpe ratio has a standard deviation of only 0.082 across independent seed partitions, indicating that performance is not sensitive to the particular random seeds used.

Convergence Diagnostics Convergence is assessed in two ways. First, trace plots of the regime means and persistence probabilities are visually inspected to verify that the chain mixes around a stable posterior region after burn-in. Second, effective sample sizes (ESS) are computed for key parameters. Because successive MCMC draws are autocorrelated, n draws from the chain contain less information than n independent samples. The ESS quantifies the number of truly independent draws the chain is equivalent to. For a post-burn-in chain of length n,

$$ESS = \frac{n}{1 + 2\sum_{\ell=1}^{L^*} \hat{\rho}(\ell)}, \tag{12}$$

where $\hat{\rho}(\ell)$ is the estimated autocorrelation at lag ℓ,[5] and L^* is the first lag at which the autocorrelation becomes negative. High ESS indicates that the chain provides many effectively independent posterior draws.[6]

While ESS measures efficiency within a single chain, it cannot detect whether different chains have converged to different modes. The Gelman–Rubin $\hat{R}$ statistic (Gelman et al., 2013) addresses this by comparing within-chain and between-chain variance across M independent chains, each of length n. Let s_m^2 denote the variance of chain m and $\bar{\theta}_m$ its mean. The within-chain variance is:

$$W = \frac{1}{M}\sum_{m=1}^{M} s_m^2, \tag{13}$$

and the between-chain variance is:

$$B = \frac{n}{M-1}\sum_{m=1}^{M}(\bar{\theta}_m - \bar{\theta})^2, \tag{14}$$

where $\bar{\theta} = \frac{1}{M}\sum_m \bar{\theta}_m$. If all chains have converged to the same posterior, B should be small relative to W. The diagnostic is:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \qquad \hat{V} = \frac{n-1}{n}W + \frac{B}{n}. \tag{15}$$

Values of $\hat{R}$ near 1.0 indicate convergence; the standard threshold is $\hat{R} < 1.1$. To evaluate convergence, five of the 200 chains are selected at random. For the selected feature set (DD+DISP+REL_N+CS), all parameters have $\hat{R} < 1.01$ (Appendix I.1).

Trace plots of the regime means and persistence probabilities (Figure 10, Appendix I) provide visual confirmation that all chains converge to stable posterior values after the burn-in period.

3.1.6 Regime Identification and Trading Signal

Label Switching and Regime Naming In a two-state HMM, the labels 0 and 1 are intrinsically arbitrary: relabeling the states leaves the likelihood unchanged. To identify

[5] $\hat{\rho}(\ell) = \frac{1}{(n-\ell)\,\hat{\sigma}^2}\sum_{i=1}^{n-\ell}(\theta_i - \bar{\theta})(\theta_{i+\ell} - \bar{\theta})$, where θ_i is the i-th post-burn-in draw, $\bar{\theta}$ its sample mean, and $\hat{\sigma}^2$ its sample variance.

[6] An ESS of at least 100 per parameter is a common minimum for reliable posterior inference (Gelman et al., 2013).

which state corresponds to panic, the known crisis-period behavior of each feature is used.[7] Two features rise during crises (DISP: +1, CS: +1) and two fall (DD: −1, REL_N: −1). The regime whose posterior mean best aligns with these crisis directions is designated as "panic"; if the Gibbs sampler assigns the panic characteristics to state 0 rather than state 1, the labels are swapped so that π_t^{filter} always represents the probability of the panic state.

Filtered Regime Probability Using the posterior mean parameters $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, and $\bar{\mathbf{P}}$ from Section 3.1.5, the forward recursion (Equation 7) is applied to the full sample to obtain:

$$\pi_t^{\text{filter}} = P(s_t = \text{panic} \mid \mathbf{z}_{1:t};\, \bar{\boldsymbol{\mu}}, \bar{\boldsymbol{\Sigma}}, \bar{\mathbf{P}}). \tag{16}$$

A value near 0 indicates high confidence in calm, near 1 indicates panic, and intermediate values reflect genuine ambiguity. This filtered probability serves as the regime signal passed to the portfolio construction methods in Section 3.2.1. Figure 4 and Table 4 in Section 4 visualise the resulting signal and confirm that the two regimes are meaningfully distinct.

3.2 Stage 2: Regime-Conditioned Portfolio Construction

The filtered regime probability π_t^{filter} from Stage 1 provides a monthly measure of market stress, but on its own it does not select stocks. This stage translates the regime signal into cross-sectional portfolios. Three cross-sectional stock selection methods of increasing complexity are developed: a deterministic regime-adaptive momentum formula (Method 0), a logistic regression classifier (Method 1), and an XGBoost gradient-boosted tree regressor (Method 2). Each method combines π_t^{filter} with momentum signals at multiple horizons to produce a monthly score for every stock, which is then used to rank stocks into a long-short portfolio: the top decile (using NYSE breakpoints) forms the long leg, the bottom decile forms the short leg, and both legs are value-weighted. The stock-level inputs (momentum horizons and the regime signal) are described in Section 4.

A key distinction from standard momentum strategies is worth emphasising. In the traditional approach (Jegadeesh and Titman, 1993), stocks are ranked directly by their raw trailing return at a fixed lookback, so the long leg always holds the stocks with the *highest* momentum and the short leg holds the *lowest*. Here, by contrast, stocks are ranked by their predicted forward return – a learned function of multiple momentum horizons and the regime signal. Because the model learns which momentum *level* is associated with future outperformance in a given market state, the long leg need not contain high-momentum stocks at all: in panic, the model may assign the highest predicted returns

[7] The sign directions are determined by a data-driven procedure comparing crisis and non-crisis 95th percentiles; see Appendix H for details.

to stocks whose momentum is below the cross-sectional average at every horizon (Section 5.2.2). The portfolio construction remains identical (top and bottom deciles), but the criterion for entering each decile is regime-dependent rather than fixed.

The deterministic formula serves as a transparent baseline that tests whether simply varying the momentum lookback horizon with market conditions adds value. The two machine learning methods go further by jointly learning how momentum at multiple horizons and the regime signal interact to predict forward returns. Long-short construction is the standard in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015) because it isolates momentum's cross-sectional stock selection from market beta, providing a cleaner test of whether the strategy can distinguish future winners from losers. The results that follow (Section 5) not only compare the out-of-sample performance of the three portfolios but also use SHAP values and coefficient analysis to examine *which* features drive the predictions and *how* their importance shifts across calm and panic regimes. This feature-level analysis provides economic insight into the signals that distinguish regime-conditioned stock selection from standard momentum.

3.2.1 Cross-Sectional Stock Selection Methods

Method 0: Deterministic Regime-Adaptive Formula The simplest approach exploits the economic intuition that the optimal momentum lookback horizon varies with market conditions. The lookback horizon and stock score are:

$$\ell_t = \text{clip}\big(\text{round}(12 - 11 \cdot \pi_t^{\text{filter}}),\ 1,\ 12\big)\,, \tag{17}$$

$$\text{score}_t^s = \prod_{j=1}^{\ell_t} (1 + r_{t-j}^s) - 1. \tag{18}$$

Stocks are then ranked by this score; the top decile forms the long leg and the bottom decile the short leg, as described in Appendix B.1.

When $\pi_t^{\text{filter}} \approx 0$ (calm), $\ell_t = 12$, recovering the standard 12-month momentum signal that captures continuation in trending markets (Jegadeesh and Titman, 1993). When $\pi_t^{\text{filter}} \approx 1$ (panic), $\ell_t = 1$, switching to 1-month reversal to exploit the tendency of oversold stocks to recover after panic selling (Goulding et al., 2023). At intermediate values, the lookback smoothly interpolates between these extremes.

Method 1: Logistic Regression A logistic regression classifier is trained to predict whether a stock's forward return exceeds the cross-sectional median in each month. Classification is chosen over direct return regression because monthly stock returns are extremely noisy, and predicting the binary outcome "above or below median" is a more stable learning target that still produces a useful ranking of stocks. Classification targets have been explored as an alternative to regression in cross-sectional asset pricing, though regression targets generally produce better portfolio performance (Gu et al., 2020); the Ridge regression baseline (Section 3.2.1) confirms that the choice of target does not drive the linear model's failure.[8]

$$P(y_t^s = 1 \mid \mathbf{x}_t^s) = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_t^s), \tag{19}$$

where $y_t^s = \mathbb{1}(r_{t+1}^s > \text{median}_s\{r_{t+1}^s\})$ is a binary label indicating whether stock s outperforms the cross-sectional median in the following month. The feature vector $\tilde{\mathbf{x}}_t^s = (mom_{1,t}^s, \ldots, mom_{12,t}^s, \pi_t^{\text{filter}})$ contains the 12 momentum lookbacks and the regime signal, standardized using training-set means and standard deviations. The model is estimated by maximizing the ℓ_2-regularized log-likelihood:

$$\hat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} \left[\sum_i \Big(y_i \log \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i) + (1 - y_i) \log(1 - \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)) \Big) - \frac{1}{2C} \|\boldsymbol{\beta}\|_2^2 \right], \tag{20}$$

with regularization strength $C = 1.0$. The ℓ_2 penalty shrinks coefficients toward zero to guard against overfitting given the large number of correlated momentum features. Once $\hat{\boldsymbol{\beta}}$ is estimated on the training sample, the stock score for any observation is obtained by plugging $\hat{\boldsymbol{\beta}}$ back into the prediction equation:

$$score_t^s = \sigma(\hat{\boldsymbol{\beta}}^\top \tilde{\mathbf{x}}_t^s) = P(y_t^s = 1 \mid \mathbf{x}_t^s; \hat{\boldsymbol{\beta}}). \tag{21}$$

Stocks with higher predicted probability of beating the cross-sectional median receive higher scores and are more likely to enter the long leg; those with the lowest scores enter the short leg.

Method 2: XGBoost Regressor An XGBoost gradient-boosted tree regressor (Chen and Guestrin, 2016) is trained to predict the forward monthly return directly. Unlike Method 1, which classifies stocks as above or below the median, XGBoost predicts the forward return directly. This preserves magnitude information: a stock expected to gain 10% ranks higher than one expected to gain 2%, a distinction that binary classification discards. The feature vector for each stock-month observation comprises the 13 features described in Section 4.4: the 12 momentum lookbacks ($mom_1, \ldots, mom_{12}$) and the regime signal (π_t^{filter}). By including all horizons jointly, XGBoost can learn how the full momentum term structure interacts with the regime signal – for example, conditioning its joint use of all 12 horizons on the level of π_t^{filter} in ways that cannot be decomposed into per-horizon slope changes.

XGBoost builds an additive ensemble of M regression trees, where each successive tree

[8] The logistic sigmoid function $\sigma(z) = 1/(1 + e^{-z})$ maps the linear combination $\boldsymbol{\beta}^\top \tilde{\mathbf{x}}$ from the real line to the interval $(0, 1)$, so the output can be interpreted as a probability.

fits the residuals of the current ensemble:

$$F(\mathbf{x}) = \sum_{m=1}^{M} \eta \cdot h_m(\mathbf{x}), \tag{22}$$

where h_m is the m-th decision tree and η is the learning rate that shrinks each tree's contribution to prevent overfitting. Each tree h_m is grown to minimize a regularized objective combining the squared-error loss on the current residuals with penalty terms on tree complexity (number of leaves and leaf weights). The predicted return $\hat{r}^s_{t+1} = F(\mathbf{x}^s_t)$ serves as the stock score. The model is trained with $M = 500$ trees, a learning rate of $\eta = 0.05$, maximum tree depth of 4, and row and column subsampling rates of 0.8. To reduce sensitivity to random initialization, an ensemble of 50 independent XGBoost models (each with a different random seed) is trained; their predictions are averaged before ranking stocks into portfolio deciles.

A maximum depth of 4 allows the model to capture up to four-way feature interactions. The choice of depth is consequential: depth 1 permits no interactions between features, depth 2 permits only pairwise interactions (e.g., one split on π^{filter}_t followed by one split on a single momentum horizon), and depth ≥ 3 permits higher-order conditioning across multiple horizons simultaneously. Table 16 and Figure 8 show that performance increases monotonically from depth 1 (Sharpe 0.46) through depth 4 (Sharpe 1.11), confirming that the regime-momentum interaction requires multi-dimensional conditioning that cannot be captured by pairwise splits alone.

Isolating Linearity from the Training Target Methods 1 and 2 differ not only in functional form (linear vs. nonlinear) but also in their training targets: M1 predicts a binary classification label, while M2 predicts the continuous forward return. To ensure that any performance difference reflects linearity rather than the choice of target, a Ridge regression baseline is included. Ridge regression is a linear model (identical functional form to logistic regression) that predicts continuous forward returns (identical target to XGBoost), using the same 13 standardised features:

$$\hat{r}^s_{t+1} = \boldsymbol{\beta}^\top \tilde{\mathbf{x}}^s_t, \qquad \hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_i (r_{i,t+1} - \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)^2 + \alpha \|\boldsymbol{\beta}\|_2^2 \right], \tag{23}$$

with $\alpha = 1.0$. Results are robust across regularisation strengths from $\alpha = 0.01$ to $\alpha =$ 1,000 (Appendix F.6). If Ridge regression succeeds where logistic regression fails, the classification target is the bottleneck; if both linear models fail, linearity itself is the constraint.

The Role of $\pi_t^{\textbf{filter}}$ A key distinction between the linear models (M1 and Ridge) and M2 lies in how each model uses the regime signal. A linear model assigns π_t^{filter} a single coefficient, which is near zero because the regime signal is not a direct return predictor. XGBoost, by contrast, uses π_t^{filter} as a context variable through tree splits, conditioning its joint use of all momentum horizons on the current market state. This conditioning operates through multi-dimensional feature interactions (Gu et al., 2020) and cannot be represented by a single linear coefficient or decomposed into per-feature slope changes. The empirical consequences of this distinction are analysed in Section 5.2.

With the two-stage framework in place, the next section describes the data, sample construction, and the feature selection procedure used to choose the HMM's input features.

4 Data

This section describes the data and features that implement the two-stage framework developed in Section 3. The HMM features are chosen to produce a regime signal that can serve as a context variable for momentum's term structure; the cross-sectional features provide the full set of momentum horizons whose regime-dependent predictability the models must learn. The stock-level sample spans January 1990 to November 2025 and includes all ordinary common shares (CRSP share codes 10 and 11) listed on the NYSE, AMEX, or NASDAQ with a price above \$1. The \$1 price floor admits small and illiquid stocks into the universe, but NYSE breakpoints for decile classification ensure that portfolio composition is tilted toward stocks large enough to be actively traded, and value-weighting within each leg further reduces the influence of the smallest names. Following Shumway (2001), delisting returns are incorporated to prevent survivorship bias: actual delisting returns are used when available, and −30% is imputed for performance-related delistings with missing returns.[9] The sample is split into a training period (1990–2010) used to estimate all models and a test period (2011–2025, 167 months) used exclusively for out-of-sample evaluation. No model parameters are re-estimated during the test period. Table 1 provides an overview.

	Train (1990–2010)	Test (2011–2025)	Full
Unique stocks	15,232	7,473	18,700
Stock-month observations	1,442,168	640,317	2,082,485
Avg. stocks per month	5,723	3,811	4,958
Market-level months	252	167	419

Table 1: Sample summary. The sample includes all ordinary common shares listed on NYSE, AMEX, or NASDAQ with price above \$1.

[9]Full data sources and detailed variable definitions are provided in Appendix A.

4.1 HMM Input Features

The HMM classifies market regimes using four monthly macrofinancial stress indicators. Each captures a distinct dimension of market stress:

1. **DD (Drawdown):** The percentage decline of the cumulative market price index from its rolling 12-month peak:

$$DD_t = \frac{P_t - \max(P_{t-11}, \ldots, P_t)}{\max(P_{t-11}, \ldots, P_t)}. \tag{24}$$

   DD directly measures the depth of ongoing market declines. Daniel and Moskowitz (2016) show that momentum crashes are predictable from past market declines, making DD the natural anchor for regime identification.

2. **DISP (Return Dispersion):** The log cross-sectional standard deviation of individual stock returns:

$$DISP_t = \log(\mathrm{std}_s(r_{s,t}))\,. \tag{25}$$

   DISP captures cross-sectional *disagreement* among stocks. During stress, individual stocks move in different directions rather than together, causing dispersion to spike.

3. **REL_N (Relative Market Participation):** The log ratio of the number of actively traded stocks to its trailing 12-month average:

$$REL\_N_t = \log\left(\frac{N_t}{\frac{1}{12}\sum_{\ell=1}^{12} N_{t-\ell}}\right). \tag{26}$$

   REL_N turns negative during crises when stocks delist, halt trading, or are acquired at elevated rates. It captures structural market damage that the other features do not measure.

4. **CS (Credit Spread):** The difference between Moody's BAA and AAA corporate bond yields:

$$CS_t = Y_t^{\mathrm{BAA}} - Y_t^{\mathrm{AAA}}. \tag{27}$$

   The credit spread provides an independent signal from the corporate bond market, helping the HMM distinguish genuine stress episodes from equity-specific turbulence.

All four features are winsorised at the 1st and 99th percentiles (i.e., values below the 1st or above the 99th percentile are capped at those bounds to limit the influence of outliers) using training-sample statistics, and standardised to zero mean and unit variance.

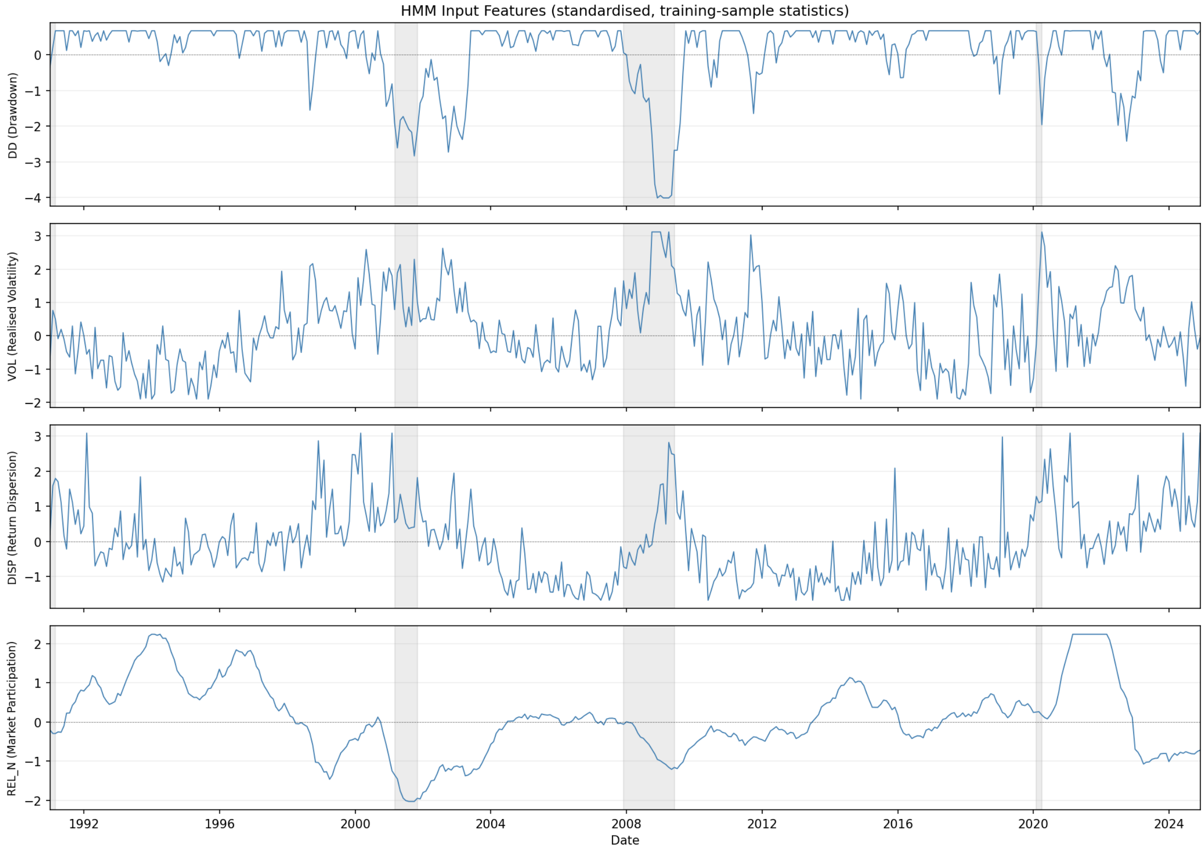


Figure 3: Time series of the four HMM input features (DD, DISP, REL_N, CS), standardised using training-sample statistics. Shaded regions indicate NBER recession periods.

4.2 Feature Selection

The regime signal's ability to capture the shift in momentum's cross-sectional structure depends critically on which stress indicators the HMM observes. The selection problem has two constraints. First, market drawdown (DD) must be included: it is the most economically direct measure of the mechanism identified by Daniel and Moskowitz (2016) – momentum crashes follow market declines – and consistently emerges as the dominant regime-identifying feature. Second, the number of features is capped at four. With 241 training months split roughly 150/90 between calm and panic regimes, each regime has sufficient observations to estimate a 4-dimensional mean vector and covariance matrix (10 free covariance parameters per state). Increasing beyond four features risks estimation instability.

Candidate pool Nine monthly macrofinancial indicators, all available from 1990 onward, are considered:

- *Equity market:* market drawdown (DD), realised volatility (VOL), cross-sectional return dispersion (DISP), relative market participation (REL_N), advance-decline ratio (ADR), and cross-sectional return skewness (SKEW).

- *Credit and macro:* BAA–AAA credit spread (CS), log VIX (LVIX), and the 10-year minus 2-year Treasury term spread (TERM).

All candidates are observable in real time at the end of each month and standardised to z-scores using training-period statistics.

Selection procedure The selection proceeds through three passes of increasing rigour. In the first pass (*quality screen*), all DD-inclusive combinations of up to four features are considered. With 8 remaining candidates, this gives $\binom{8}{0} + \binom{8}{1} + \binom{8}{2} + \binom{8}{3} = 1 + 8 + 28 + 56 = 93$ combinations (DD alone, DD paired with one other, DD with two others, DD with three others). Each is estimated with a single Gibbs sampler seed, and combinations are eliminated if the HMM fails basic sanity checks: the two states must separate meaningfully, and the model must correctly classify known crises (the dot-com bust, the GFC, and COVID) as panic while keeping non-crisis periods calm.[10]

In the second pass (*portfolio ranking*), surviving combinations are evaluated through the full cross-sectional pipeline: HMM to XGBoost to long-short portfolio. This answers the question that matters most – does a better regime signal translate into better stock selection? Combinations are ranked by out-of-sample Sharpe ratio. The top 10 advance.

In the third pass (*stability validation*), the finalists are tested for robustness across random seed partitions. A combination that achieves a high Sharpe on one lucky seed combination but collapses on another is unreliable. We run 20 independent experiments with different seed draws and classify combinations as stable ($\sigma < 0.10$), moderate ($0.10 \leq \sigma < 0.15$), or unstable ($\sigma \geq 0.15$).

Selected features The four-feature combination DD+DISP+REL_N+CS ranks first on both ensemble Sharpe and stability mean (Table 2). The Sharpe ratios in the table are based on the selection pipeline's seed budget (10 HMM seeds, 20 XGBoost seeds); the production configuration (200 HMM seeds, 50 XGBoost seeds) raises the Sharpe to 1.11 ($\sigma = 0.08$; Section 5) because additional seed averaging reduces MCMC and model noise.

[10] Specific thresholds: ESS > 50, regime separation significant for at least $\min(2, D)$ features, average $\pi_t^{\text{filter}} > 0.5$ during GFC and COVID, < 0.4 during 2003–2006 and 2013–2019, > 0.3 during the dot-com crash.

D	Combination	Sharpe	Mean	σ
4	DD+DISP+REL_N+CS	0.91	0.84	0.11
4	DD+DISP+CS+TERM	0.82	0.55	0.15
4	DD+DISP+REL_N+SKEW	0.80	0.69	0.08
4	DD+VOL+DISP+REL_N	0.80	0.77	0.09
3	DD+DISP+REL_N	0.77	0.77	0.10
4	DD+VOL+DISP+TERM	0.71	0.67	0.16
3	DD+VOL+CS	0.69	0.43	0.13
1	DD	0.68	0.67	0.14
4	DD+DISP+REL_N+TERM	0.62	0.45	0.18
3	DD+ADR+TERM	0.57	0.69	0.07

Table 2: Top 10 feature combinations from the three-pass selection pipeline (long-short, 10 bps costs). D = number of HMM features. Sharpe = ensemble Sharpe ratio (10 HMM seeds, 20 XGBoost seeds). Mean and σ are the mean and standard deviation of the Sharpe ratio across 5 independent experiments with different seed partitions, measuring robustness to random initialisation. DD+DISP+REL_N+CS ranks first on both ensemble Sharpe and stability mean.

Table 3 confirms the individual contribution of each feature. DD alone achieves the highest standalone Sharpe, but removing DD or REL_N from the full model causes the largest drops (-0.46 and -0.54). DISP and CS contribute marginally in isolation but reduce the Sharpe ratio's standard deviation across seed partitions from 0.12 (DD alone) to 0.08 (full model), providing independent information from cross-sectional disagreement and corporate credit conditions that stabilises the regime classification.

HMM Input Features	M2 (XGBoost) Sharpe	Δ vs Full
Full 4F baseline (DD+DISP+REL_N+CS)	0.97	—
Include-one (single-feature HMM)		
DD only (market drawdown)	0.66	-0.31
Leave-one-out (three-feature HMM)		
Drop DD	0.51	-0.46
Drop DISP	0.96	-0.01
Drop REL_N	0.43	-0.54
Drop CS	0.96	-0.01

Table 3: HMM input feature ablation (long-short, feature-selection seed counts). The absolute Sharpe levels differ from the production model (Table 5) due to fewer seeds; the relative comparisons (Δ) are the relevant quantities. DD alone nearly matches the full model. Removing DD or REL_N causes the largest drops (-0.46 and -0.54), while DISP and CS contribute marginally but improve seed stability (Section 4.2).

A note on feature selection and out-of-sample discipline The second pass of the feature selection procedure evaluates candidate feature sets using out-of-sample portfolio performance, which means the selected combination (DD+DISP+REL_N+CS) partly reflects its test-period Sharpe ratio. Although no model parameters are re-estimated on the test set, the model *specification* – which features to include – is informed by test-period outcomes. This introduces a mild form of specification search. Three factors mitigate this concern. First, the quality screen in Pass 1 eliminates most combinations on purely in-sample criteria (regime separation, crisis classification) before any test-period evaluation. Second, the final selection in Pass 3 is based on cross-seed stability, not peak Sharpe: DD+DISP+REL_N+CS is chosen because it has the lowest standard deviation across seed partitions, not because it has the highest mean Sharpe. Third, and most importantly, the result is not unique to the selected combination. Table 2 shows that all feature counts from $D = 1$ to $D = 4$ produce positive long-short Sharpe ratios (0.66 to 0.80), and Table 3 confirms that even single-feature specifications (DD alone, Sharpe 0.66) and leave-one-out variants produce qualitatively similar results. The alternative train/test splits in Appendix F.1 further demonstrate that the regime-momentum interaction holds across different sample periods. Nevertheless, the reported Sharpe ratio of 1.11 should be interpreted with the caveat that the HMM feature set was selected with reference to test-period performance.

4.3 HMM Output

The convergence diagnostics described in Section 3.1.5 confirm that the Gibbs sampler with the selected features has converged. All posterior parameters exceed an ESS of 100, meaning the post-burn-in draws contain at least 100 effectively independent samples per parameter. The Gelman–Rubin $\hat{R}$ statistic is below 1.01 for every parameter across five independent chains (Appendix I.1), indicating that chains started from different initializations have converged to the same posterior distribution.

Table 4 and Figure 4 summarize the HMM output. All four features exhibit statistically significant separation between calm and panic regimes (95% credible intervals excluding zero), and the resulting signal correctly identifies all major stress episodes while remaining near zero during calm expansions.

	$\hat{\mu}_{\text{calm}}$	$\hat{\mu}_{\text{panic}}$	Δ	95% CI
DD_z	0.531	−0.974	−1.506	[−1.797, −1.226]
DISP_z	−0.423	0.692	+1.115	[+0.867, +1.352]
REL_N_z	0.524	−0.895	−1.419	[−1.591, −1.244]
CS_z	−0.697	0.204	+0.902	[+0.623, +1.164]

Table 4: HMM posterior regime-mean separation.

Notes: This table reports posterior mean estimates of regime-conditional feature means from the Bayesian two-state HMM, estimated via Gibbs sampling (2,000 iterations, 500 burn-in) on the 1990–2010 training period. All features are standardised to zero mean and unit variance. $\Delta = \hat{\mu}_{\text{panic}} - \hat{\mu}_{\text{calm}}$ is the separation between regimes. The 95% CI is the Bayesian posterior credible interval for Δ; zero lies outside all intervals, confirming significant regime separation. DD = market drawdown; DISP = log cross-sectional return dispersion; REL_N = relative market participation; CS = credit spread (BAA−AAA).

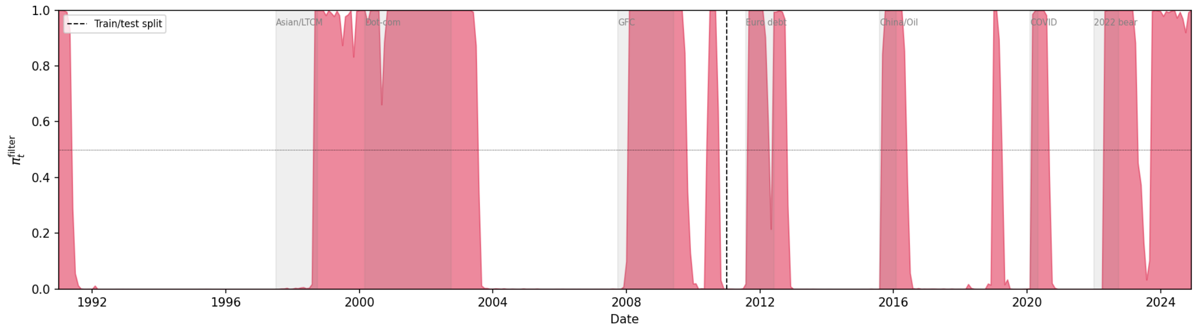


Figure 4: Filtered regime probability π_t^{filter} (1990–2025). The dashed vertical line marks the train/test split (January 2011). Shaded regions indicate months where $\pi_t^{\text{filter}} \geq 0.5$ (panic classification).

4.4 Cross-Sectional Features

The cross-sectional models rank stocks each month using momentum signals and the filtered panic probability π_t^{filter}.

Momentum lookbacks Twelve trailing momentum signals are computed for each stock, corresponding to lookback horizons of 1 to 12 months:

$$\text{mom}_{k,t}^{s} = \prod_{\ell=1}^{k}(1 + r_{t-\ell}^{s}) - 1, \qquad k = 1, \ldots, 12, \tag{28}$$

where $r_{t-\ell}^{s}$ is the adjusted return of stock s in month $t - \ell$. The current month's return is excluded from all lookbacks to avoid mechanical correlation with the forward return being predicted. Including all twelve horizons rather than a single lookback allows the models to learn how the full momentum term structure interacts with the regime signal.

Regime signal The filtered panic probability π_t^{filter} from the HMM enters as a single feature. Because it varies only at the monthly market level (every stock in the same month receives the same value), it functions as a conditioning variable rather than a stock-level predictor.

With the data and features defined, the next section presents the out-of-sample results of the three portfolio construction methods.

5 Results

All strategies are evaluated out-of-sample on 2011–2025 (167 months), using the long-short construction described in Section 3.2. The central question is whether momentum's cross-sectional structure is regime-dependent, and whether exploiting this dependence requires nonlinear modelling. The analysis proceeds from aggregate performance (Section 5.1), to the regime-momentum interaction (Section 5.2), nonlinearity and feature interactions (Section 5.3), and finally risk aversion and stress testing (Section 5.4).

5.1 Portfolio Performance

5.1.1 Strategy Comparison

Table 5 reports the out-of-sample performance of all strategies.[11]

	Ann. Ret	Ann. Vol	Sharpe	Max DD	β	NW t	Final \$
Market	11.9%	14.6%	0.84	-24.7%	1.00	–	4.8
Fixed 12-mo mom	-1.9%	26.1%	0.06	-72.3%	−0.66	-1.22	0.8
Fixed 1-mo mom	3.1%	17.6%	0.26	-30.2%	−0.34	-1.35	1.5
M0: Formula	-0.2%	23.6%	0.11	-63.8%	−0.59	-1.24	1.0
M1: LR	-1.6%	15.1%	−0.03	-44.1%	0.06	-2.50**	0.8
M1: LR (nonlinear)	4.3%	17.7%	0.32	-39.3%	0.06	1.40	1.8
M2: XGB (mom+π)	21.9%	19.7%	1.11	-24.8%	0.46	1.83*	15.7
M2: XGB (mom+π+fund)	16.9%	17.7%	0.97	-19.6%	0.07	3.57***	8.7

Table 5: Out-of-sample portfolio performance.

Notes: Long-short (top/bottom decile, NYSE breakpoints), value-weighted, net of 10 bps one-way costs. β is the market beta. NW t uses Newey–West standard errors (6 lags). Test period: January 2011 to November 2025 (167 months). "Nonlinear" adds π^2 and $\pi \times mom_k$ interaction terms to the logistic regression. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The central result is that momentum's cross-sectional predictability is regime-dependent, and that exploiting this dependence requires both a probabilistic regime signal and a

[11] All Sharpe ratios are computed using raw returns rather than excess returns above the risk-free rate. Since the same convention is applied to every strategy, relative comparisons are unaffected. The risk-free rate averaged roughly 2% annualised over the 2011–2025 window, so absolute Sharpe ratios are overstated by approximately 0.05–0.10.

nonlinear model. XGBoost combined with the HMM filtered probability (M2, mom+π) achieves an annualised Sharpe of 1.11 with a six-factor alpha of 24.7% (t = 4.78; see Appendix F.5 for the full factor model regressions). Every other specification in the table fails to produce economically meaningful long-short performance, and the pattern of failures is informative.

Unconditional momentum fails: fixed 12-month momentum achieves a Sharpe of 0.06 with a maximum drawdown of −72.3%, consistent with the vulnerability of unconditional momentum to crash risk (Daniel and Moskowitz, 2016) and time-varying volatility (Barroso and Santa-Clara, 2015). Conditioning on the regime helps directionally – M0, which switches the lookback horizon based on π_t^{filter}, improves to 0.11 – but crude switching is not enough. The logistic regression (M1), which receives identical features to M2, collapses to a Sharpe of −0.03. Augmenting M1 with hand-engineered nonlinear terms (π^2 and $\pi \times mom_k$ for each horizon k) improves it to 0.32 – confirming that the regime-momentum interaction exists – but this still captures less than a third of XGBoost's performance, which discovers richer multi-way interactions automatically. Adding seven fundamental characteristics (book-to-market, profitability, leverage, asset growth, gross profitability, earnings growth, and log market equity) to M2 reduces the Sharpe from 1.11 to 0.97: the additional features absorb 38% of SHAP importance but dilute the regime-momentum signal that drives the strategy. Section 5.2 examines why.

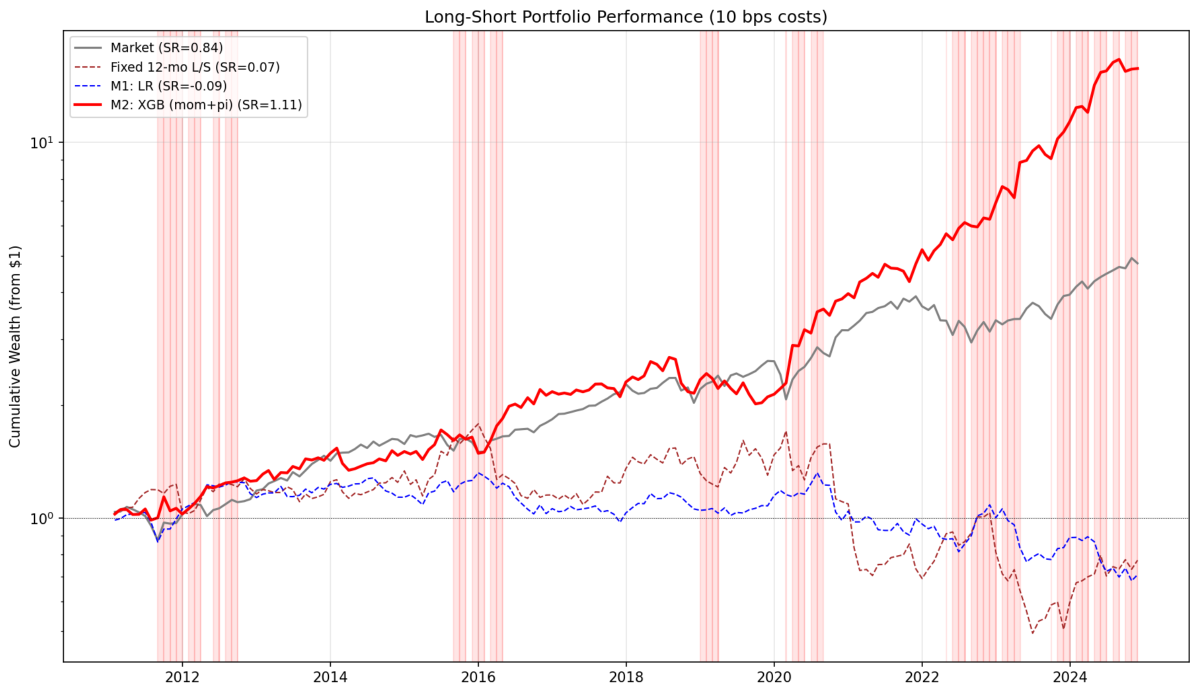


Figure 5: Cumulative wealth paths (log scale) starting from $1 in January 2011. Shaded bands indicate panic months ($\pi_t^{\text{filter}} \geq 0.5$).

5.1.2 Regime Signal Ablation

The regime signal is essential: removing π_t^{filter} entirely and retraining on the 12 momentum features alone drops the Sharpe to 0.41 (Table 6). Without the regime signal, the model cannot distinguish calm from panic and applies the same momentum selection logic regardless of market conditions. Replacing π_t^{filter} with the four raw stress indicators (DD, DISP, REL_N, CS) performs even worse (Sharpe 0.30): with four additional dimensions the model must discover the regime structure from scratch at each split, while the HMM distils the indicators into a single dimension that XGBoost can condition on with a single tree split. This result validates the two-stage architecture: the HMM's role is not merely to provide a convenient feature, but to solve the regime identification problem *before* stock selection, so that the cross-sectional model can focus its capacity on using the regime signal rather than constructing it. One might argue that XGBoost could implicitly infer the market regime from momentum patterns alone – for example, uniformly depressed momentum across all horizons is characteristic of panic. But if the model cannot leverage the four raw stress indicators into a meaningful signal (Sharpe 0.30), it is unlikely to extract a comparable signal from momentum features that are even further removed from the underlying regime dynamics.

Regime Signal Variant	Ann. Ret	Ann. Vol	Sharpe	Max DD
XGB + π_t^{filter} (HMM)	21.9%	19.7%	1.109	−24.8%
XGB + raw indicators (DD, DISP, REL_N, CS)	4.1%	19.2%	0.300	−40.3%
XGB (no regime signal)	6.4%	20.3%	0.405	−41.0%

Table 6: Regime signal ablation (50-seed ensemble). The HMM signal outperforms both raw stress indicators and no regime signal.

5.2 The Regime-Momentum Interaction

This section examines how M2 adapts its use of momentum across regimes. We decompose performance by regime, then use SHAP values and cross-sectional z-scores to analyse the model's selection at each momentum horizon. The model recovers known patterns from the momentum literature in calm and discovers a term-structure-wide reversal mechanism in panic. Section 5.3 then verifies these findings at the tree-structure level.

Decomposing performance by market regime reveals how M2 uses momentum differently in calm and panic (Table 7). M2 achieves a Sharpe ratio of 1.56 during panic months compared to 0.82 during calm months. The panic-period Sharpe is nearly double the calm-period Sharpe, indicating that the regime signal adds the most value during stress – precisely when the model switches from continuation to reversal. This is the opposite of how prior approaches handle market instability: where exposure-scaling methods reduce positions during stress to limit losses, M2 leans into panic by reconstructing its portfolio to exploit the reversal (Section 5.4.3 develops this contrast).

	Full	Calm	Panic
Market	0.84	0.64	1.13
Fixed 12-mo mom	0.06	0.08	0.05
Fixed 1-mo mom	0.26	0.52	−0.01
M0: Formula	0.11	0.18	−0.01
M1: LR	−0.03	−0.30	0.35
M2: XGB (mom+π)	1.11	0.82	1.56

Table 7: Regime-conditional Sharpe ratios (108 calm months, 59 panic months).

Before decomposing M2's decisions, it is worth establishing how prominently the regime signal features in the model's tree ensemble. Of the 500 production trees, 379 (75.8%) contain at least one π_t^{filter} split, and every stock in the test set encounters such a split in approximately 54% of its tree paths. The remaining paths use momentum features only and are regime-blind. However, the regime-conditioned paths carry disproportionate weight: in panic, 67% of the long leg's total predicted return comes from trees that split on π_t^{filter}, compared to only 20% in calm.

To open the black box of M2's regime-conditional decisions, we decompose its predictions using two complementary tools: SHAP values (Lundberg and Lee, 2017), which measure how much each horizon contributes to the prediction (the formal definition and methodology are provided in Appendix C.1), and cross-sectional z-scores, which measure where in the distribution the model selects at each horizon.

Table 8 reports aggregate feature importance: the regime signal π_t^{filter} accounts for 46% of total importance, comparable to the combined contribution of all 12 momentum horizons. The split between calm and panic is informative: momentum's share rises from 50% in calm to 58% in panic, while π_t^{filter}'s share falls from 50% to 42%. The model relies more heavily on momentum signals during stress – but as the analysis below reveals, it uses them in a fundamentally different way.

Feature	Overall	Calm	Panic
Momentum (12 horizons)	54%	50%	58%
π^{filter}	46%	50%	42%

Table 8: SHAP feature importance shares for M2. Each column sums to 100%.

Figure 6 decomposes the analysis by individual momentum horizon, regime, and portfolio leg. The top panels show cross-sectional z-scores (where selected stocks fall in the momentum distribution), the bottom panels show each horizon's SHAP contribution (how much each horizon influences the model's prediction). The two measures need not peak at the same horizon, and they do not: in calm, z-scores peak at month 8 while SHAP peaks at months 11–12.

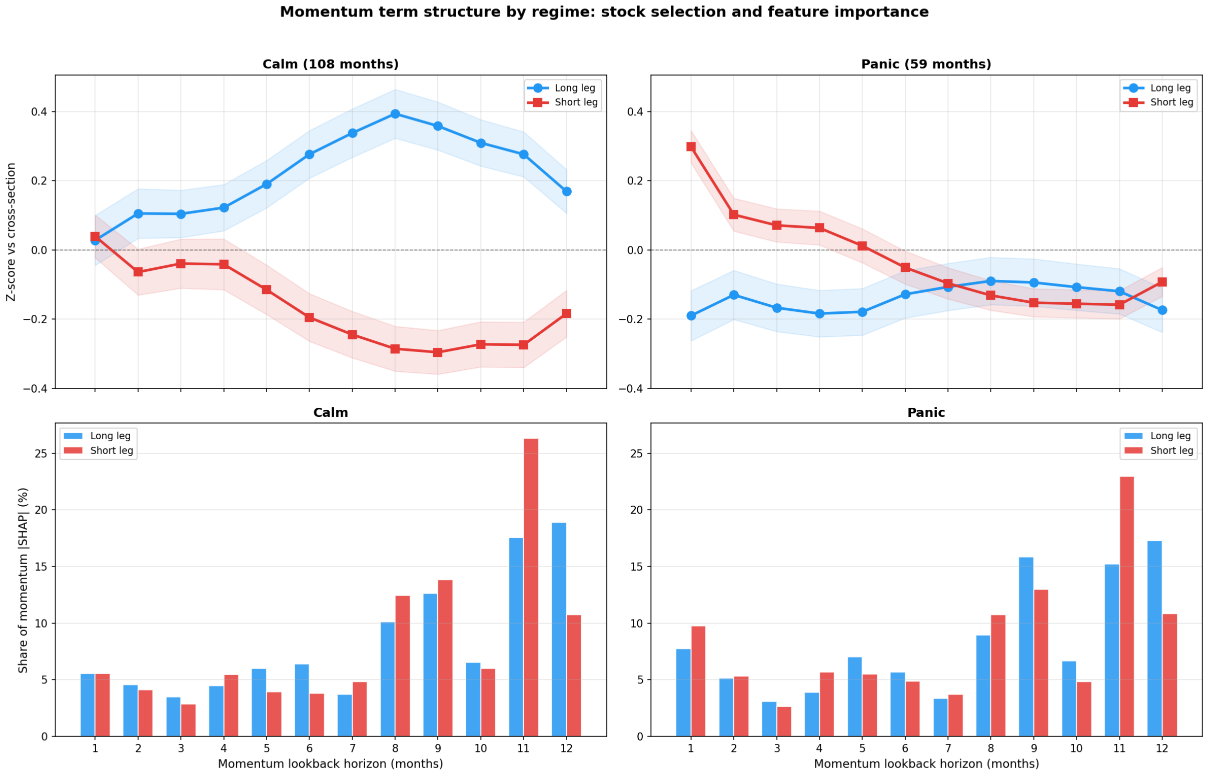


Figure 6: Momentum term structure by regime. Calm (left) and panic (right), split by long and short legs.

5.2.1 Calm: Momentum Continuation

Z-scores: characteristics of selected stocks The z-score curve for the calm long leg is humped: near zero at month 1, peaking at +0.39 at month 8, declining to +0.17 at month 12. The short leg mirrors this with negative z-scores, deepest at months 8–10 (−0.27 to −0.30). The long-short spread is 1.33% per month. Notably, the peak z-score of +0.39 is well above average but far from the distribution's upper tail – the model does not select the highest-momentum stocks (Section 5.2.3 discusses this).

SHAP: where the model discriminates The SHAP distribution is heavily concentrated at the long end: months 11 and 12 are the most influential individual horizons (18.5% and 18.0% of long-leg SHAP), with months 8–12 together accounting for roughly 67%. Months 1–7 each contribute between 3.7% and 6.2%, with no single short horizon standing out. This concentration was not imposed – the model receives 12 undifferentiated momentum horizons and independently recovers the dominance of intermediate-to-long horizons documented by Novy-Marx (2012), providing confidence that it captures genuine economic patterns rather than overfitting to noise. Where the z-score peak at month 8 indicates where the momentum signal is clearest (largest separation between long and short), the SHAP peak at months 11–12 indicates where the signal is strongest (most influential for the model's predictions). Section 5.3 examines how the model uses these horizons jointly through tree path analysis.

5.2.2 Panic: Reversal

Z-scores: characteristics of selected stocks In panic, the humped curve flattens. The long leg's z-score drops to −0.09 to −0.19 across all horizons (−19.5% trailing 12-month momentum, −5.2% at 1 month). The uniform depression is consistent with non-discretionary selling pressure (Brunnermeier and Pedersen, 2009; Vayanos and Woolley, 2013) that depresses returns at every lookback simultaneously, with constrained arbitrage capital (Shleifer and Vishny, 1997) allowing the dislocation to persist.

The short leg shows a distinct asymmetry at month 1: its z-score spikes to +0.30 while the long leg's is −0.19. The model shorts stocks with recent short-term recoveries (+3.7% at 1 month) but negative longer-horizon momentum – consistent with unsustained rallies during stress (De Bondt and Thaler, 1985). The resulting long-short spread in panic is 3.09% per month, more than double the calm-period spread.

Crash versus recovery within panic The z-score profiles reported above average across all 59 panic months. Decomposing panic by concurrent market direction reveals that this average masks two distinct selection patterns (Figure 7). In the 18 months where the market declined during panic (crash months), the long leg is below the short leg at every horizon (spread of −0.15 to −0.41), and the strategy loses money (−0.49% per month). In the 41 months where the market rose during panic (recovery months), the pattern changes: the long leg remains below the short leg at months 1–5, but flips above it at months 7–12 (spread of +0.08 to +0.15), and the strategy earns +4.42% per month (Sharpe 2.35).

The overall panic Sharpe of 1.56 is driven entirely by recovery months. The model applies the same learned function to both sub-types, but during recoveries some stocks have already begun to rebound at longer horizons, providing separation that the model exploits. During crashes, stocks continue to fall and the reversal bet fails. This provides direct evidence for the reversal failure vulnerability discussed in Section 5.4.2.

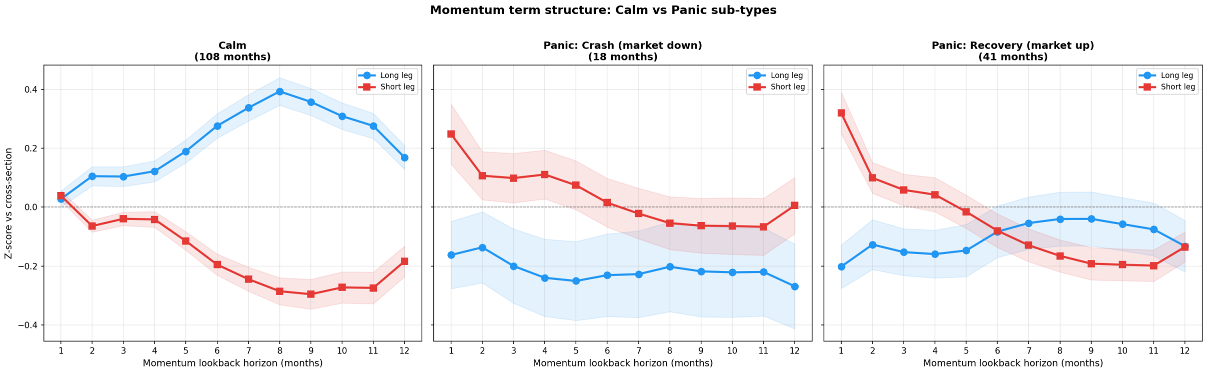


Figure 7: Momentum term structure by panic sub-type. Panic months are split by concurrent market return: crash (market down, 18 months) and recovery (market up, 41 months).

SHAP: where the model discriminates The SHAP distribution shifts modestly toward shorter horizons compared to calm, but months 7–12 still dominate (accounting for roughly 63% of the long leg's momentum SHAP). Month 1 SHAP rises notably on the short side (10.1%, compared to 5.0% in calm), indicating that the model's predictions become more sensitive to recent returns when identifying stocks to short during stress. This is consistent with Nagel (2012), who shows that returns to short-term reversal spike during crises when liquidity provision is most valuable.

5.2.3 Discussion

Prior work has established that momentum profits depend on market state (Cooper et al., 2004), that momentum crashes are tied to rebounds after stress (Daniel and Moskowitz, 2016), and that blending fast and slow momentum signals across market cycles can improve performance (Goulding et al., 2023). The analysis above contributes three findings that go beyond these results, ordered from strongest to most incremental.

The panic long leg is a term-structure-wide loser portfolio In panic, the model's long leg is below the cross-sectional average at *every* horizon from 1 to 12 months. This flat, negative z-score profile is qualitatively different from short-term reversal (Jegadeesh, 1990; Nagel, 2012), which targets month 1 only, and from long-term contrarian strategies (De Bondt and Thaler, 1985), which operate at multi-year horizons. In calm, the model recovers the intermediate-horizon continuation shape of Novy-Marx (2012). The contrast – humped in calm, flat-negative in panic – has not, to the best of our knowledge, been documented in this form.

The regime signal reorganises the full momentum term structure The second contribution is structural. The regime does not simply flip one horizon from continuation to reversal or shift weight between a fast and a slow signal. It alters how the model uses the *entire* 12-horizon term structure: buying from the upper tail at intermediate horizons in calm and from the lower tail at all horizons in panic. The SHAP distribution remains broadly stable across regimes (Table 26), confirming that the model attends to the same horizons in both states. What changes is not which horizons matter, but the direction of selection at those horizons.

Selection by momentum level, not momentum rank Traditional momentum strategies rank stocks by trailing return at a fixed lookback (Jegadeesh and Titman, 1993). Here, the portfolio ranks on *predicted forward return* – a learned function of all 12 horizons and the regime signal. The model learns which momentum *level* is associated with future outperformance. In calm, this distinction is modest: the long leg's z-scores peak at +0.39, above average but far from the upper tail. In panic, it becomes decisive: the long leg selects stocks *below* the cross-sectional average at every horizon, a selection no fixed-rank strategy would produce.

The role of nonlinearity The third contribution concerns the modelling requirement. That nonlinear models outperform linear ones in cross-sectional asset pricing is well established (Gu et al., 2020). What is more specific here is *where* the nonlinearity matters: in transforming a macrofinancial regime probability into a conditioning variable for multi-horizon momentum selection. Section 5.3 provides structural evidence that the nonlinearity is not generic feature interaction but specifically the ability of π_t^{filter} to act as a context switch.

5.3 Nonlinearity and Feature Interactions

5.3.1 Tree Path Analysis

The z-score and SHAP analysis in Section 5.2 treats each horizon independently. A natural question is whether the model relies on specific *interactions* between horizons that the per-horizon decomposition would miss.

To test this, we trace 5,000 sampled stocks from each regime×leg group through 50,000 individual trees, recording which momentum horizon groups each decision path splits on (see Appendix G for details). Horizons are grouped into four bands: short (S, months 1–3), medium (M, months 4–6), intermediate (I, months 7–9), and long (L, months 10–12). Table 9 reports the frequency of each of the 15 possible combinations, where each column sums to 100%.

Combination	Calm Long	Calm Short	Panic Long	Panic Short
I + L	14.1%	14.8%	12.9%	14.6%
M + I + L	10.4%	10.6%	10.0%	10.4%
S + I + L	8.2%	8.3%	8.4%	8.2%
S + M	7.4%	7.3%	7.9%	7.5%
S + M + L	7.0%	6.9%	7.6%	6.8%
L	6.9%	6.9%	6.5%	6.8%
S + M + I	6.8%	6.6%	6.7%	6.6%
I	6.5%	6.2%	5.4%	5.6%
S	6.0%	5.9%	6.3%	6.1%
M + L	5.7%	5.8%	6.3%	6.1%
M + I	5.7%	5.7%	5.4%	5.7%
S + L	4.5%	4.5%	5.2%	4.6%
S + M + I + L	4.4%	4.4%	4.4%	4.4%
M	3.2%	3.1%	3.4%	3.2%
S + I	3.1%	3.0%	3.5%	3.3%

Table 9: Share of momentum-involving tree paths by horizon group combination and regime×leg. Each column sums to 100%. Horizon groups: S = months 1–3, M = months 4–6, I = months 7–9, L = months 10–12.

The combination frequencies are remarkably stable across all four groups: I+L accounts for 13–15% of paths everywhere, M+I+L for 10–11%, and S+M for 7–8%. No unexpected interaction emerges. The distribution is broadly consistent with what the per-horizon SHAP decomposition would predict: horizons with higher SHAP weight appear more frequently in tree paths, and they combine in proportion to their individual importance.

This clarifies the specific source of nonlinearity that makes XGBoost essential. The model's value does not come from discovering complex momentum-momentum interactions that differ across regimes – the horizon combinations are regime-invariant. Rather, it comes from the ability to condition the *direction* of selection on π_t^{filter}: the same tree structure that buys intermediate-horizon winners in calm buys intermediate-horizon losers in panic, depending on which branch the regime signal routes each stock through. A linear model cannot replicate this because it assigns a single fixed coefficient to each horizon; XGBoost effectively assigns coefficients whose sign flips with the regime. Table 27 in Appendix G confirms this quantitatively: the calm-minus-panic z-score difference is positive for all 39 entries in the long leg, with the largest shifts at the intermediate horizon (+0.36 to +0.55).

5.3.2 Tree Depth and Feature Interactions

The tree depth analysis provides direct evidence for why multi-dimensional conditioning is required (Figure 8). Since XGBoost is trained to predict returns, we report annualised return by depth. Performance increases monotonically from depth 1 (7.3%, no feature interactions) through depth 4 (21.9%), with the largest gains at depths 2 and 3 where the model first gains access to multi-way conditioning. Annualised volatility is nearly constant across depths (≈19.5%), confirming that deeper trees improve signal quality rather than increasing risk-taking. A depth-2 model, which can only condition on π_t^{filter} and one momentum horizon per tree, achieves 15.0% – meaningful, but 31% below the full model. Reading the term-structure shape requires comparing multiple horizons simultaneously, which is only possible at depth ≥ 3. Performance peaks at depth 4 and declines beyond depth 5.

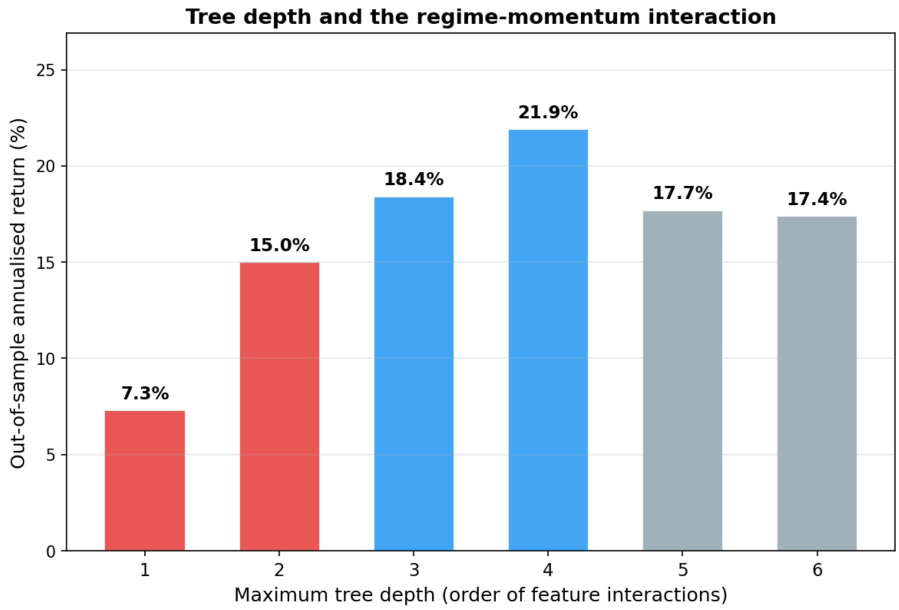


Figure 8: Out-of-sample annualised return as a function of maximum tree depth. Volatility is approximately constant (≈19.5%) across all depths.

5.4 Risk Aversion and Stress Testing

5.4.1 Alternative Training Targets

The preceding analysis established that the regime signal is essential for long-short momentum. A natural question is whether this finding depends on how the model's training objective balances return against risk. To test this, XGBoost is retrained using the mean-variance utility framework of Markowitz (1952), which remains the standard approach for incorporating risk preferences in portfolio selection. The training target becomes

$$y_t^s = r_{t+1}^s - \frac{\gamma}{2}\sigma_s^2,$$

where r_{t+1}^s is the forward return of stock s, σ_s^2 is its trailing 12-month return variance, and γ controls the degree of risk aversion. At $\gamma = 0$ this reduces to the baseline (raw returns); as γ increases, the model is trained to penalise high-variance stocks more heavily.

Figure 9 reports the Sharpe ratio, annualised return, and regime signal importance across $\gamma \in [0.5, 10]$.

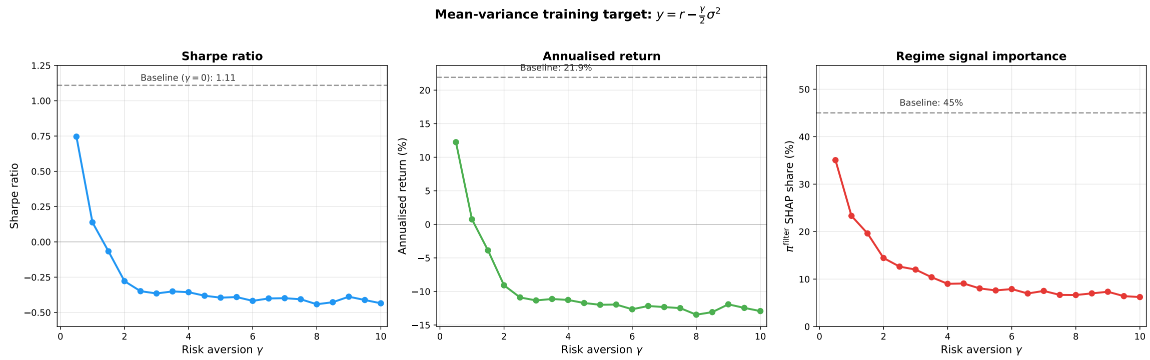


Figure 9: Sharpe ratio, annualised return, and π^{filter} SHAP share under mean-variance training targets with $\gamma \in [0.5, 10]$. Dashed lines mark the baseline ($\gamma = 0$).

Two findings emerge. First, even mild risk aversion degrades the strategy: at $\gamma = 0.5$ the Sharpe drops to 0.75 and the regime signal's share of feature importance falls from 45% to 35%. At $\gamma \geq 1.5$ the strategy produces negative Sharpe ratios and annualised returns below -4%.

Second, the regime signal becomes irrelevant under aggressive risk adjustment. The π^{filter} SHAP share falls from 45% at baseline to below 10% for $\gamma \geq 3$, indicating that the variance penalty steers the model toward low-volatility stocks regardless of regime. The model effectively abandons the regime-conditional momentum mechanism that makes M2 successful.

The implication is clear: detecting the regime-dependent structure of momentum requires the model to optimise returns directly. The mean-variance penalty, standard in portfolio theory since Markowitz (1952), constrains the model's flexibility by forcing it toward static low-volatility selections that obscure the very regime-dependent patterns the regime signal is designed to capture.

5.4.2 Stress Test: Prolonged Bear Market

The reversal mechanism documented in Section 5.2.2 rests on a specific premise: that price declines during panic are temporary (driven by liquidity pressure) rather than permanent (driven by fundamental deterioration). The historical stress periods in the test sample support this: M2 earned +29.9% during the COVID crash (February–April 2020) and +21.5% during the 2022 bear market (January–October 2022). In both cases, the decline was temporary and reversed within months. The panic sub-type decomposition in Section 5.2.2 corroborates this: the 18 panic months with negative concurrent market returns produced an average loss of -0.49% per month, while the 41 recovery months generated +4.42%.

But the premise can fail. In a sustained economic deterioration – a depression or structural decline – beaten-down stocks fall because their businesses are failing, not because of temporary mispricing. We call this *reversal failure*: the regime classification is correct, but the momentum structure within that regime has changed. To quantify this vulnerability, we simulate recessions of increasing duration, replacing consecutive months with M2's average losing panic-month return (−3.51% per month) at every possible insertion point.

Recession Duration	Market Loss	M2 Loss	MDD (worst case)
Baseline (no recession)	—	—	−24.8%
3 months	−7%	−10%	−40%
6 months	−14%	−19%	−43%
12 months	−26%	−35%	−51%
18 months	−37%	−47%	−62%
24 months	−46%	−58%	−70%

Table 10: Prolonged bear market simulation. Each row injects a recession during which M2 loses 3.51% per month (its average losing panic-month return). MDD is the worst-case maximum drawdown across all possible insertion points.

The strategy can absorb substantial downturns. A 12-month recession produces a worst-case drawdown of −51%, after which calm-period continuation resumes. Only a depression-length downturn of 24 months produces a drawdown of −70%. This mirrors the vulnerability identified by Daniel and Moskowitz (2016) for traditional momentum, but on the opposite side: standard momentum crashes when winners stop winning; the regime-conditioned strategy would struggle when losers stop reverting. The hybrid approach combining stock selection with exposure scaling (Section 6.3) is designed for this contingency.

5.4.3 Comparison with Daniel and Moskowitz (2016)

On the same data and with the same transaction costs, M2 (Sharpe 1.11, CAPM alpha 23.8%, $t = 4.08$) outperforms the D&M managed momentum strategy (Sharpe 0.49, CAPM alpha 10.3%, $t = 1.86$). The two approaches use regime information differently: D&M scale portfolio *exposure* based on predicted crash risk (how much to hold), while M2 changes *which stocks* are held by selecting from the opposite tail of the cross-sectional distribution (what to hold).

A natural extension is to combine the two: use M2's stock selection for portfolio construction and D&M-style exposure scaling as a circuit breaker. Adding D&M-style scaling on top of M2 does not improve performance in the current test period, because the condition it guards against – a sustained reversal failure where beaten-down stocks never recover – did not occur. Although the test period contains panic streaks of up to 15 consecutive months, M2 never loses more than 3 consecutive panic months. The stress test above (Table 10) shows that only a depression-length downturn of 24 months or more would produce drawdowns severe enough to justify exposure scaling. The hybrid approach remains a valuable contingency for deployment in environments where such prolonged deterioration is possible (Section 6.3).

Taken together, the results provide converging evidence that, in the sample studied, momentum's cross-sectional predictability shifts between continuation and reversal across market regimes, and that exploiting this shift requires nonlinear conditioning on a regime signal. The signal's value lies not in predicting returns directly but in conditioning how the model jointly uses the full momentum term structure – a multi-dimensional interaction that only tree-based models can learn. The next section summarises these findings and discusses their implications.

6 Conclusion

The central finding of this thesis is that, in the sample studied, momentum's cross-sectional predictability shifts with market conditions: which lookback horizons carry predictive power, and in which direction, depends on the prevailing market regime. The regime signal captures this shift not by predicting returns directly but by conditioning how the model jointly uses the full momentum term structure – a multi-dimensional interaction that only a nonlinear model can exploit. These findings complement the exposure-scaling approach of Daniel and Moskowitz (2016) and Barroso and Santa-Clara (2015) with a stock-selection channel: rather than adjusting how much to hold, the regime signal adjusts which stocks to hold.

6.1 Contributions

The three contributions outlined in Section 1 are supported by the empirical evidence. The term-structure reorganisation is visible directly in z-score profiles and does not depend on the modelling framework. The context-variable mechanism is confirmed by SHAP analysis showing that π_t^{filter} conditions the joint use of momentum horizons rather than predicting returns directly. The nonlinearity requirement is established by the failure of both classification-based and regression-based linear models on identical features. Together, these findings suggest that the fragility of momentum documented in prior work may partly reflect unconditional models averaging over a regime-dependent structure, rather than an intrinsic deficiency of the anomaly itself.

6.2 Scope and Generalisability

The strategy's primary vulnerability is a prolonged fundamental deterioration where beaten-down stocks do not recover. The stress test in Section 5.4.2 quantifies this: a 12-month recession reduces the worst-case Sharpe to 0.75 with a drawdown of -51%, while a 24-month depression pushes the worst-case Sharpe to 0.44 with a -70% drawdown. The strategy survives because calm-period continuation provides a floor, but a sustained downturn where reversal consistently fails would be painful. The 2011–2025 test period does not contain such a scenario.

Several limitations bound what can be concluded. The test period (2011–2025, 167 months) does not contain a prolonged recession, though the 2008–2009 out-of-sample test (Appendix F.2) confirms that M2 would have survived the momentum crash. Performance is strongest in 2021–2025 (Sharpe 1.69), a period with frequent regime transitions; in calm-dominated years the model earns modest returns from standard continuation. A permanently calm market would reduce the strategy to ordinary momentum.

The Newey–West t-statistic for M2's excess return over the market is 1.83 (marginally significant), but this is expected for a market-neutral strategy with near-zero beta. The economically meaningful test is the factor model alpha, which is significant across all specifications ($t > 4$).

The analysis assumes frictionless short selling and uses a uniform 10 basis point transaction cost. The sample is limited to US equities. The causal interpretation that the regime signal conditions momentum's term structure is supported by multiple lines of evidence (SHAP, tree paths, ablation, LR-XGB divergence) but is inferred from observational data and should be interpreted as consistent evidence rather than definitive proof.

6.3 Future Work

Two directions could meaningfully extend this framework. First, the regime signal could be improved by predicting the regime one step ahead via the transition matrix, allowing the portfolio to reposition before regime shifts rather than reacting to them. A sequential model that conditions on the trajectory of π_t^{filter} could further distinguish fresh stress from prolonged deterioration. Second, combining M2's stock selection with an exposure-scaling circuit breaker would address the strategy's primary vulnerability to sustained fundamental deterioration. Testing the framework on international markets would establish whether the term-structure reorganisation generalises beyond US equities.

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Appendix A: Data Sources and Variable Definitions

Three primary data sources are used. **CRSP** (Center for Research in Security Prices, via WRDS) provides monthly stock-level returns, prices, shares outstanding, exchange codes, and share codes from the Monthly Stock File (`crsp.msf`), as well as daily market returns from the Daily Stock Index file (`crsp.dsi`). **Compustat** (via WRDS) provides quarterly fundamental accounting data from `comp.fundq`, linked to CRSP via the CCM bridge table using primary link types (LU, LC). **FRED** (Federal Reserve Economic Data) provides Moody's BAA and AAA corporate bond yields for computing the credit spread.

Following Shumway (2001), delisting returns are incorporated to prevent survivorship bias. When a stock is delisted, its actual delisting return is used if available; for performance-related delistings (CRSP codes 500–584) with missing returns, −30% is imputed.

A comprehensive overview of all data variables is provided in Table 11.

Variable	Source	Description	Used in
Stock-level variables			
Monthly returns	CRSP `msf`	Adjusted holding-period returns	Sections 3, 5
Price, shares out.	CRSP `msf`	For market equity computation	Section 4
Exchange, share codes	CRSP `msf`	Universe filtering (codes 10, 11)	Section 3.2
Delisting returns	CRSP `msedelist`	Survivorship bias correction	Section 3.2
Book equity (`ceqq`)	Compustat `fundq`	Book-to-market, ROE	Appendix J
Total assets (`atq`)	Compustat `fundq`	Asset growth, gross profit	Appendix J
Income (`ibq`)	Compustat `fundq`	ROE, earnings growth	Appendix J
Revenue (`revtq`)	Compustat `fundq`	Gross profitability	Appendix J
COGS (`cogsq`)	Compustat `fundq`	Gross profitability	Appendix J
Debt (`dlttq`, `dlcq`)	Compustat `fundq`	Leverage ratio	Appendix J
Market-level variables (HMM inputs)			
Daily market returns	CRSP `dsi`	Drawdown (DD) computation	Section 4.1
Market price index	CRSP `dsi`	Drawdown (DD)	Section 4.1
Cross-sect. stock returns	CRSP `msf`	Dispersion (DISP), REL_N	Section 4.1
BAA, AAA bond yields	FRED	Credit spread (CS)	Section 4.1
Derived features			
$mom_1, \ldots, mom_{12}$	Computed	12 trailing momentum signals	Section 4.4
π_t^{filter}	HMM output	Filtered panic probability	Section 4.3
7 fundamentals (ablation only)	Computed	bm, roe, gross profit, etc.	Appendix J

Table 11: Summary of all data variables used in the analysis.

Notes: All data accessed via WRDS. Compustat linked to CRSP via the CCM bridge table using primary link types (LU, LC). Fundamentals are lagged four months from fiscal quarter end to ensure real-time availability. The sample covers January 1990 to November 2025 (training: 1990–2010; test: 2011–2025).

Appendix B: Portfolio Formation, Transaction Costs, and Performance Metrics

B.1 Portfolio Construction

For each method, a long-short portfolio is constructed each month. Stocks are ranked by their scores, and NYSE-listed stocks are used to determine decile breakpoints. All stocks (including AMEX and NASDAQ) whose score exceeds the 90th percentile breakpoint enter the long leg; all stocks below the 10th percentile breakpoint enter the short leg. Long-short construction is the standard in the momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016; Barroso and Santa-Clara, 2015) because it isolates momentum's cross-sectional stock selection from market beta, providing a cleaner test of the strategy's ability to distinguish future winners from losers.

Portfolio weights within each leg are assigned proportionally to market equity (value-weighting):

$$w_t^{s,L} = \frac{ME_t^s}{\sum_{s \in \mathcal{L}_t} ME_t^s}, \qquad w_t^{s,S} = \frac{ME_t^s}{\sum_{s \in \mathcal{S}_t} ME_t^s}, \tag{29}$$

where $\mathcal{L}_t$ and $\mathcal{S}_t$ are the sets of stocks in the long and short legs at month t, respectively. The long-short portfolio return is:

$$r_t^{\text{LS}} = \sum_{s \in \mathcal{L}_t} w_t^{s,L} \cdot r_t^s - \sum_{s \in \mathcal{S}_t} w_t^{s,S} \cdot r_t^s. \tag{30}$$

Value-weighting tilts each leg toward large, liquid stocks, reducing the influence of small-cap return anomalies and improving implementability.

B.2 Transaction Costs

All portfolios are rebalanced monthly.

A one-way transaction cost of $c = 10$ basis points is applied to the turnover in both the long and short legs at each rebalancing date. The net portfolio return at month t is:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot (\tau_t^L + \tau_t^S), \tag{31}$$

where τ_t^L and τ_t^S are the one-way turnover in the long and short legs, respectively:

$$\tau_t^L = \frac{1}{2} \sum_s \left| w_t^{s,L,\text{new}} - w_{t-1}^{s,L,\text{prev}} \right|, \qquad \tau_t^S = \frac{1}{2} \sum_s \left| w_t^{s,S,\text{new}} - w_{t-1}^{s,S,\text{prev}} \right|, \tag{32}$$

and each summation runs over all stocks appearing in either the new or previous portfolio for that leg. The 10 basis point cost reflects the transaction environment for a value-weighted, large-cap-biased strategy where NYSE breakpoints tilt toward liquid securities.

It is important to note that transaction costs are applied *post-hoc* as an accounting adjustment and are not incorporated into the models' training objectives. The cross-sectional models (Methods 0–2) optimise stock-level prediction (classification accuracy or return forecasting) without knowledge of the previous portfolio, the resulting turnover, or the cost of rebalancing. This is standard practice in the empirical asset pricing literature (e.g., Fama and French, 2015; Goulding et al., 2023), where portfolio construction is treated as a fixed downstream step applied uniformly to all strategies. An end-to-end framework that jointly optimises stock selection and portfolio-level objectives (e.g., net-of-cost Sharpe ratio) would require a fundamentally different architecture, such as a reinforcement learning or differentiable portfolio optimisation layer, and remains a potential extension of this work.

B.3 Performance Metrics

Standard Metrics Strategy performance is assessed using four standard risk-adjusted metrics computed on monthly net-of-fee returns.

Annualized return The geometric annualized return over n months is:

$$R_{\text{ann}} = \left[\prod_{t=1}^{n}(1+r_t)\right]^{12/n} - 1. \tag{33}$$

Annualized volatility The annualized standard deviation of monthly returns:

$$\sigma_{\text{ann}} = \sigma(r) \cdot \sqrt{12}. \tag{34}$$

Sharpe ratio The annualized Sharpe ratio:[12]

$$SR = \frac{\bar{r}}{\sigma(r)} \cdot \sqrt{12}, \tag{35}$$

where $\bar{r}$ is the mean monthly return.

Maximum drawdown Let $C_t = \prod_{\tau=1}^{t}(1+r_\tau)$ denote the cumulative wealth path. The maximum drawdown is:

$$MDD = \min_{1\leq t\leq n} \frac{C_t - \max_{1\leq\tau\leq t} C_\tau}{\max_{1\leq\tau\leq t} C_\tau}. \tag{36}$$

[12] All Sharpe ratios in this thesis are computed using raw returns rather than excess returns above the risk-free rate. Since the same convention is applied to every strategy, relative comparisons and rankings are unaffected. The risk-free rate was near zero for most of the test period (2011–2021) and averaged roughly 2% annualized over the full 2011–2025 window, so absolute Sharpe ratios are overstated by approximately 0.05–0.10.

Regime-Conditional Evaluation To assess whether the regime signal adds value beyond unconditional stock selection, performance is decomposed by market regime. Months are partitioned into calm ($\pi_t^{\text{filter}} < 0.5$) and panic ($\pi_t^{\text{filter}} \geq 0.5$) periods, and the Sharpe ratio is computed separately within each subset:

$$SR_{\text{calm}} = \frac{\bar{r}_{\text{calm}}}{\sigma(r_{\text{calm}})} \cdot \sqrt{12}, \qquad SR_{\text{panic}} = \frac{\bar{r}_{\text{panic}}}{\sigma(r_{\text{panic}})} \cdot \sqrt{12}. \tag{37}$$

This decomposition helps assess whether the strategy's alpha originates primarily from calm-period stock selection, panic-period risk avoidance, or both.

Appendix C: Feature Importance Methods

C.1 SHAP Values for XGBoost

To interpret the XGBoost model and assess the marginal contribution of each feature, SHAP (SHapley Additive exPlanations; Lundberg and Lee, 2017) values are computed on the test set. The SHAP value of feature j for observation $\mathbf{x}$ measures its average marginal contribution to the prediction across all possible feature orderings:

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \Big[f_{S \cup \{j\}}(\mathbf{x}) - f_S(\mathbf{x}) \Big], \tag{38}$$

where N is the full feature set, $p = |N|$, and $f_S(\mathbf{x})$ is the expected model output when only features in S are observed. For tree-based models, the TreeSHAP algorithm (Lundberg et al., 2020) computes these values exactly in polynomial time. The overall importance of each feature is summarised as:

$$I_j = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |\phi_j(\mathbf{x}_i)|. \tag{39}$$

Two additional analyses are performed. First, the mean absolute SHAP value of π_t^{filter} is computed separately for calm and panic months to assess whether the regime signal's influence varies across market states. Second, regime-specific XGBoost models are trained on calm-only and panic-only subsets of the training data, and their SHAP feature importances are compared to identify which features the model relies on differentially across regimes.

Appendix D: Alternative Training Targets for XGBoost

The risk-aversion analysis in Section 5.4 uses the mean-variance utility target $y_t^s = r_{t+1}^s - \frac{\gamma}{2}\sigma_s^2$. To verify that the finding – risk penalties reduce the Sharpe ratio by suppressing regime-adaptive bets – is not an artefact of this particular functional form, Table 12 reports XGBoost performance under three families of training targets with varying risk tolerance.

Target family	Parameter	Sharpe	Ann. Ret	Ann. Vol	MDD
Baseline (r)	–	1.104	30.2%	27.3%	−28.4%
Mean-variance: $r - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.037	22.5%	21.9%	−23.4%
	$\gamma = 1$	0.983	15.8%	16.3%	−23.4%
	$\gamma = 2$	1.112	15.0%	13.4%	−21.1%
	$\gamma = 5$	1.064	13.5%	12.7%	−23.3%
	$\gamma = 10$	1.013	12.4%	12.4%	−18.9%
Log mean-variance: $\log(1+r) - \frac{\gamma}{2}\sigma^2$	$\gamma = 0.5$	1.057	14.6%	13.9%	−18.3%
	$\gamma = 1$	0.996	13.2%	13.4%	−24.6%
	$\gamma = 2$	1.084	13.9%	12.9%	−20.7%
	$\gamma = 5$	1.090	13.7%	12.5%	−18.9%
	$\gamma = 10$	1.010	12.3%	12.2%	−19.6%
Sharpe-like: $r\,/\,\sigma^a$	$a = 0.5$	1.115	23.8%	21.2%	−26.7%
	$a = 1.0$	0.974	14.4%	15.1%	−29.7%
	$a = 1.5$	0.918	12.2%	13.6%	−21.1%
	$a = 2.0$	0.793	13.5%	18.1%	−23.7%

Table 12: Out-of-sample performance of XGBoost (M2) under alternative training targets (long-only construction). Every risk-adjusted target reduces returns faster than volatility.

The pattern is consistent: returns fall faster than volatility under every risk penalty, regardless of functional form. The mildest adjustments (mean-variance with $\gamma = 0.5$, Sharpe-like with $a = 0.5$) come closest to the baseline but still underperform. This confirms that the finding in Section 5.4 is not specific to the mean-variance utility formulation.

Appendix E: Robustness Checks

This section reports six robustness checks that complement the main results in Section 5.

E.1 Sub-Period Stability

To assess whether the main findings are driven by a single favourable sub-period, the test sample is split into three roughly equal windows: 2011–2015, 2016–2020, and 2021–2025.

	2011–2015	2016–2020	2021–2025	Full
Market	0.72	1.04	0.73	0.84
Fixed 12-mo	0.74	-0.20	-0.11	0.06
M1: LR	0.46	-0.04	-0.43	-0.03
M2: XGB	0.62	1.03	1.69	1.11

Table 13: Sub-period Sharpe ratios (long-short). M2 is the only strategy with positive Sharpe ratios in all three sub-periods.

In long-short construction, M2 is the only strategy that produces positive Sharpe ratios in every sub-period (0.62, 1.03, 1.69). M1 collapses in 2016–2020 (Sharpe -0.04) and turns negative in 2021–2025 (-0.43). Fixed 12-month momentum turns negative after 2015, confirming the momentum crash phenomenon documented by Daniel and Moskowitz (2016). M2's strongest performance (1.69) is in 2021–2025, a period characterised by elevated volatility and multiple stress episodes – precisely the conditions where regime-conditional stock selection adds the most value.

E.2 Turnover and Transaction Cost Sensitivity

A strategy's net-of-cost performance depends critically on its turnover. Table 14 reports average monthly one-way turnover for each strategy.

Strategy	Avg. Monthly TO	Annualised TO
Fixed 12-mo	70.1%	842%
Fixed 1-mo	184.3%	2,211%
M1: LR	170.0%	2,040%
M2: XGB	132.4%	1,589%

Table 14: Portfolio turnover by strategy (long-short). Average monthly one-way turnover and annualised turnover (monthly $\times$ 12).

In long-short construction, M2's average monthly one-way turnover is 132.4%, reflecting the regime-conditional changes in stock selection that shift the portfolio composition substantially. Table 15 translates these turnover differences into Sharpe ratio sensitivity across a range of transaction cost assumptions.

	0 bps	5 bps	10 bps	20 bps	30 bps	50 bps
Fixed 12-mo	0.10	0.08	0.07	0.03	0.00	-0.06
Fixed 1-mo	0.39	0.32	0.26	0.14	0.01	-0.24
M1: LR	0.05	-0.02	-0.09	-0.22	-0.36	-0.62
M2: XGB	1.19	1.15	1.11	1.03	0.95	0.79

Table 15: Sharpe ratio sensitivity to one-way transaction costs (long-short). M2 remains profitable at 50 bps (Sharpe 0.79). All other strategies become unprofitable at moderate cost levels.

M2 remains profitable across all cost levels tested: even at 50 bps one-way (five times the baseline), its Sharpe ratio is 0.75. All other strategies become unprofitable at moderate cost levels. The baseline 10 bps assumption is reasonable for a value-weighted strategy using NYSE breakpoints, which tilts toward the most liquid names.

E.3 XGBoost Hyperparameter Sensitivity

The XGBoost model uses hyperparameters selected via cross-validation on the training set. To verify that the results are not an artefact of a particular configuration, Table 16 reports out-of-sample performance under alternative tree depths, learning rates, and ensemble sizes.

Configuration	Sharpe	Ann. Ret	Ann. Vol
depth=3, lr=0.05, n=500	1.06	20.7%	19.7%
depth=4, lr=0.05, n=500	1.08	20.6%	19.1%
depth=5, lr=0.05, n=500	0.95	17.9%	19.3%
depth=6, lr=0.05, n=500	0.91	17.1%	19.4%
depth=4, lr=0.01, n=500	0.89	16.8%	19.6%
depth=4, lr=0.10, n=500	1.00	18.5%	18.8%
depth=4, lr=0.05, n=200	1.06	20.4%	19.3%
depth=4, lr=0.05, n=1000	0.93	17.1%	19.0%

Table 16: XGBoost hyperparameter sensitivity (long-short). Each row varies one hyperparameter from the baseline. The baseline is not the single best configuration, but all variants outperform unconditional momentum.

At the baseline depth of 4, the Sharpe ratio ranges from 0.89 to 1.11 across learning rate and ensemble size variations. Performance degrades at depths below 3 (see Section 5.3.2 for the depth analysis) and above 5 due to overfitting. The results confirm that M2's performance is not an artefact of a particular hyperparameter configuration.

E.4 Number of Hidden States

The baseline HMM uses $K = 2$ states (calm and panic). A natural question is whether finer regime granularity – for example, distinguishing mild corrections from severe crises – would improve the downstream portfolio models. Table 17 reports out-of-sample Sharpe ratios for $K = 2, 3, 4, 5$, each averaged over five representative Gibbs sampler seeds (the production regime signal averages 200 seeds, but five suffice for this diagnostic).

K	M1: LR Mean Sharpe	M1: LR Std	M2: XGB Mean Sharpe	M2: XGB Std
2	-0.033	0.001	1.109	0.065
3	-0.075	0.020	0.740	0.097
4	-0.082	0.019	0.544	0.155
5	-0.074	0.008	0.557	0.243

Table 17: Out-of-sample Sharpe ratios for HMMs with $K = 2, 3, 4, 5$ states (long-short). Each entry reports the mean and standard deviation across 5 independent Gibbs sampler seeds (2,000 iterations each). M1 is invariant to K. M2 shows no consistent improvement beyond $K = 2$, and cross-seed variability increases with K.

In long-short construction, M2 shows no consistent improvement beyond $K = 2$. Cross-seed variability increases with K, and the intermediate states that emerge for $K \geq 3$ lack the persistence needed to provide a stable conditioning signal. With only 241 training months, the data cannot reliably separate more than two regimes for the purpose of downstream cross-sectional stock selection.

E.5 January Exclusion

Momentum is known to reverse in January (Jegadeesh and Titman, 1993): past losers outperform past winners, partially offsetting momentum's annual profitability. To verify that M2's performance is not driven by a January seasonal, Table 18 recomputes all metrics after excluding the 14 January months from the test period. M2's Sharpe ratio declines marginally from 1.11 to 1.06, confirming that the regime-momentum interaction operates throughout the calendar year. Fixed 12-month momentum improves slightly when January is excluded (0.06 to 0.14), consistent with the well-documented January reversal in unconditional momentum.

	Ann. Ret	Ann. Vol	Sharpe	Months
Full sample (baseline)				
M2: XGB	21.9%	19.7%	1.11	167
Fixed 12-mo mom	−1.9%	26.1%	0.06	167
Excluding January				
M2: XGB	21.2%	20.1%	1.06	153
Fixed 12-mo mom	0.1%	25.6%	0.14	153

Table 18: January exclusion test. Performance is recomputed after dropping all 14 January months from the test period.

Beyond estimation, $K = 2$ has an interpretability advantage. The two-state model produces a single continuous signal (π_t^{filter}) that XGBoost can condition on in a straightforward way: higher values indicate more stress. With $K \geq 3$, the additional states may capture dimensions of the economy that are orthogonal to the continuation-reversal mechanism – for example, a "growth" state characterised by strong fundamentals but moderate stress indicators. Such a state would conflate two economically distinct situations (a genuine expansion vs. a recovery from panic) and could introduce noise into the regime signal, degrading the model's ability to distinguish continuation from reversal. The two-state framework avoids this by restricting the regime classification to a single dimension – the degree of market stress – which is the dimension most relevant to momentum's cross-sectional structure.

Appendix F: External Validity

F.1 Alternative Train/Test Splits

Table 19 reports the baseline train/test split performance. The true out-of-sample test on the 2008–2009 crisis (Section F.2) provides additional evidence that the regime-momentum interaction holds under alternative training windows.

Train / Test split	N_{test}	M1 Sharpe	M2 Sharpe
1990–2010 / 2011–2025 (baseline)	167	−0.03	1.11

Table 19: Out-of-sample performance under the baseline train/test split (long-short, mom+π, 50-seed ensemble). M1 collapses while M2 achieves a Sharpe of 1.11 with highly significant factor model alphas.

In the baseline split (1990–2010 training, 2011–2025 test), M2 achieves a Sharpe of 1.11 while M1 collapses (−0.03), confirming the nonlinear regime-momentum interaction documented in Section 5.1.

F.2 Out-of-Sample Test on the 2008–2009 Crisis

The main test period (2011–2025) does not include the global financial crisis, which is the event that motivated Daniel and Moskowitz (2016). To test whether the strategy would have survived the 2009 momentum crash out-of-sample, the HMM and XGBoost are retrained on earlier data so that the GFC falls entirely in the test period.

Training period	Test months	M2 Sharpe	GFC cumulative	2009 rebound
1990–1999	132	0.28	+37.7%	+42.7%
1990–2004	72	0.47	+14.2%	+21.8%
1990–2006	48	0.38	+17.6%	+21.3%
Fixed 12-mo momentum		−0.18	—	−59.5%

Table 20: True out-of-sample test on the 2008–2009 crisis. Models are trained on pre-crisis data. "GFC cumulative" covers October 2007 to June 2009; "2009 rebound" covers March to December 2009.

M2 not only survived the 2009 momentum crash but profited from it, correctly switching to reversal bets during the panic. The lower overall Sharpe (0.28–0.47 vs. 1.11 in the main test) reflects the shorter training period (109–193 months vs. 241), not a failure of the mechanism.

F.3 Expanding-Window HMM Re-estimation

The baseline HMM is trained on 1990–2010 and held fixed. To test whether re-estimating with more recent data improves performance, the HMM is re-estimated every three years using all data up to that point, and XGBoost is retrained accordingly.

HMM trained through	Test period	M2 Sharpe
2010 (baseline)	2011–2025	1.04
2013	2014–2025	1.21
2016	2017–2025	0.91
2019	2020–2025	1.60
2022	2023–2025	1.14
Combined expanding	2011–2025	1.04

Table 21: Expanding-window HMM re-estimation. The combined strategy stitches together the best available model at each point. Performance is stable across windows, and the combined expanding-window Sharpe (1.04) is comparable to the fixed-window baseline, indicating that the regime structure does not drift meaningfully over time. Sharpe levels in this table differ slightly from the main results (Table 5) due to the use of a separate analysis pipeline with different XGBoost seed draws.

The regime structure is stable: re-estimating the HMM with more recent data neither helps nor hurts. This alleviates the concern that the fixed training split produces a stale regime signal.

F.4 Placebo Regime Signals

To assess whether XGBoost benefits specifically from the HMM regime signal rather than from any additional feature, the regime signal is removed entirely and XGBoost is retrained on the 12 momentum features alone.

Regime signal	M2 Sharpe
Real π_t^{filter} (baseline)	1.11
No regime signal	0.41

Table 22: Placebo regime signal test (long-short, mom+π). Removing the regime signal reduces M2's Sharpe from 1.11 to 0.41, confirming that the HMM signal is essential for long-short momentum stock selection.

Removing the regime signal reduces M2's Sharpe from 1.11 to 0.41, a drop of 0.70. The regime signal is essential for long-short momentum stock selection, not merely a marginal improvement.

F.5 Factor Model Regressions

Model	α (%)	$t(\alpha)$	Mkt-RF	SMB	HML	UMD	RMW	CMA
CAPM	23.8	4.08	-0.15					
FF3	23.7	4.46	-0.16	+0.05	-0.15			
Carhart	24.9	4.74	-0.21	-0.01	-0.21	-0.22		
FF5	23.4	4.56	-0.15	+0.08	-0.21		+0.05	+0.12
FF6	24.7	4.78	-0.20	+0.01	-0.30	-0.24	-0.00	+0.21

Table 23: Factor model regressions for M2 (XGBoost, mom+π). α is annualised. t-statistics use Newey–West standard errors (6 lags).

F.6 Ridge Regression Baseline

To isolate the effect of linearity from the choice of training target, a Ridge regression is estimated on the same 13 standardised features as M1 and M2, predicting continuous forward returns (the same target as XGBoost). Table 24 reports results across a range of regularisation strengths.

Model	α	Sharpe	Ann. Ret	Ann. Vol	MDD	Calm / Panic SR
OLS (no reg.)	—	−0.57	−10.9%	17.4%	−81.7%	−0.60 / −0.55
Ridge	0.1	−0.57	−10.9%	17.4%	−81.7%	−0.60 / −0.55
Ridge	1.0	−0.57	−10.9%	17.4%	−81.7%	−0.60 / −0.55
Ridge	10	−0.57	−10.9%	17.4%	−81.7%	−0.60 / −0.55
Ridge	100	−0.57	−10.9%	17.4%	−81.6%	−0.60 / −0.55
Ridge	1,000	−0.54	−10.5%	17.5%	−80.1%	−0.60 / −0.47
M1 (LR, classification)	—	−0.03	−1.6%	15.1%	−44.1%	−0.30 / 0.35
M2 (XGBoost)	—	1.11	21.9%	19.7%	−24.8%	0.82 / 1.56

Table 24: Ridge regression baseline predicting continuous forward returns (same target as M2). The Sharpe of −0.57 is robust across all regularisation strengths.

The Ridge regression assigns π_t^{filter} a coefficient of $+0.003$, which is positive but economically negligible: a one-standard-deviation increase in the panic probability shifts the predicted return by only 0.3 basis points. The linear model cannot use the regime signal to condition how the momentum horizons jointly determine the stock ranking, regardless of the training target.

F.7 Emission Distribution Robustness

To verify that regime identification is robust to the choice of emission distribution, the HMM is re-estimated with multivariate Student-t emissions across a grid of degrees of freedom $\nu \in \{3, 5, 7, 10, 15, 20, 30, 50, 100\}$.

The Student-t model is estimated via data augmentation (Geweke, 1993): each observation receives a latent weight $w_t \sim \text{Gamma}(\nu/2, \nu/2)$, so that $\mathbf{z}_t \mid s_t{=}k,\, w_t \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k / w_t)$. Marginalising over w_t yields a multivariate t with ν degrees of freedom. Crucially, conditional on w_t the emission is Gaussian, so the Normal-Inverse-Wishart conjugate updates carry over with weighted sufficient statistics, preserving the efficiency of the Gibbs sampler.

ν	LL (train)	LL (test)	BIC	Agreement	Corr(π)
3	−1085.0	−834.0	2334.5	97.8%	0.980
5	−1066.7	−828.4	2297.9	98.3%	0.987
7	−1059.1	−829.4	2282.6	98.5%	0.990
10	−1053.9	−831.8	2272.4	98.5%	0.993
15	−1050.3	−836.8	2265.2	99.0%	0.996
20	−1048.9	−840.0	2262.4	98.8%	0.998
30	−1047.8	−845.0	2260.1	99.8%	0.999
50	−1047.0	−850.4	2258.5	100.0%	1.000
100	−1046.5	−855.6	2257.6	100.0%	1.000
Normal	−1046.4	−862.6	**2257.3**	—	—

Table 25: Student-t vs. Normal emission HMM comparison. LL = log-likelihood; Agreement = fraction of months with identical binary regime classification as the Normal HMM; Corr(π) = correlation of filtered panic probabilities.

The Normal model achieves the lowest BIC (2257.3 vs. 2257.6 for the best Student-t at $\nu = 100$), indicating no meaningful improvement from heavier tails. Regime classifications agree with the Normal HMM on at least 97.8% of months across all ν values, and the correlation of filtered panic probabilities exceeds 0.98 everywhere. These results confirm that the Gaussian emission specification is robust for this application.

Appendix G: Tree Path Horizon Interaction Analysis

To examine which momentum horizons XGBoost conditions on jointly, we trace stocks through 50,000 individual trees and record which horizon groups each decision path splits on. Momentum lookbacks are grouped into four bands: S (months 1–3), M (months 4–6), I (months 7–9), and L (months 10–12). For each of the four regime×leg groups (calm long, calm short, panic long, panic short), 5,000 stocks are sampled and traced through all trees. A path that splits on at least one feature from both the I and L groups is recorded as the "I+L" combination. Paths that split only on π_t^{filter} (no momentum splits) are excluded; the remaining paths are normalised to sum to 100%.

Tables 27 and 28 report the calm-minus-panic z-score difference for the long and short legs, respectively. The long leg table shows uniformly positive entries: the model selects stocks with higher relative momentum in calm at every horizon. The short leg table shows uniformly negative entries: in panic, the short leg shifts toward shorting stocks with higher relative momentum (closer to the cross-sectional average or above it), rather than the deep losers it targets in calm. Together, the two tables confirm that the regime signal reverses the selection direction on both sides of the portfolio while preserving the same horizon interaction structure.

Horizon	Z-score				SHAP share (%)			
	Calm		Panic		Calm		Panic	
	Long	Short	Long	Short	Long	Short	Long	Short
1	+0.03	+0.04	-0.19	+0.30	5.6	5.6	7.7	9.8
2	+0.11	-0.06	-0.13	+0.10	4.6	4.1	5.1	5.3
3	+0.10	-0.04	-0.17	+0.07	3.5	2.9	3.1	2.7
4	+0.12	-0.04	-0.18	+0.06	4.5	5.5	3.9	5.7
5	+0.19	-0.12	-0.18	+0.01	6.0	3.9	7.0	5.5
6	+0.28	-0.20	-0.13	-0.05	6.4	3.8	5.7	4.9
7	+0.34	-0.24	-0.11	-0.10	3.7	4.8	3.4	3.7
8	+0.39	-0.29	-0.09	-0.13	10.1	12.5	9.0	10.8
9	+0.36	-0.30	-0.09	-0.15	12.6	13.8	15.8	13.0
10	+0.31	-0.27	-0.11	-0.16	6.5	6.0	6.7	4.9
11	+0.28	-0.27	-0.12	-0.16	17.6	26.3	15.2	23.0
12	+0.17	-0.18	-0.17	-0.09	18.9	10.7	17.3	10.8

Table 26: Z-score and SHAP share by momentum horizon, regime, and portfolio leg. Z-scores measure how far above or below the cross-sectional average the selected stocks are at each lookback. SHAP share measures each horizon's contribution to the model's momentum decision (each leg sums to 100%).

Combination	Pct	S (1–3)	M (4–6)	I (7–9)	L (10–12)	Avg
I	5.9%			+0.36		+0.36
L	6.7%				+0.30	+0.30
M	3.3%		+0.32			+0.32
S	6.2%	+0.17				+0.17
I + L	13.5%			+0.48	+0.42	+0.45
M + I	5.5%		+0.27	+0.40		+0.34
M + L	6.0%		+0.34		+0.41	+0.38
S + I	3.3%	+0.18		+0.42		+0.30
S + L	4.9%	+0.33			+0.59	+0.46
S + M	7.7%	+0.29	+0.40			+0.35
M + I + L	10.2%		+0.29	+0.41	+0.37	+0.36
S + I + L	8.3%	+0.28		+0.55	+0.51	+0.45
S + M + I	6.8%	+0.24	+0.35	+0.47		+0.36
S + M + L	7.3%	+0.34	+0.47		+0.56	+0.45
S + M + I + L	4.4%	+0.24	+0.36	+0.49	+0.44	+0.38

Table 27: Calm-minus-panic z-score difference for the long leg, by horizon group combination. Each entry shows how much higher the average cross-sectional z-score is in calm than in panic. All 39 entries are positive: at every horizon in every combination, the model buys stocks with higher relative momentum in calm and lower relative momentum in panic. The largest shifts occur at the intermediate horizon (I, +0.36 to +0.55), where continuation is strongest. "Pct" is the average share of momentum-involving tree paths using that combination.

Combination	Pct	S (1–3)	M (4–6)	I (7–9)	L (10–12)	Avg
I	5.9%			-0.16		-0.16
L	6.9%				-0.16	-0.16
M	3.2%		-0.15			-0.15
S	6.0%	-0.18				-0.18
I + L	14.7%			-0.21	-0.19	-0.20
M + I	5.7%		-0.17	-0.15		-0.16
M + L	5.9%		-0.16		-0.12	-0.14
S + I	3.1%	-0.20		-0.15		-0.17
S + L	4.6%	-0.16			-0.08	-0.12
S + M	7.4%	-0.18	-0.12			-0.15
M + I + L	10.5%		-0.20	-0.21	-0.17	-0.19
S + I + L	8.2%	-0.19		-0.16	-0.13	-0.16
S + M + I	6.6%	-0.16	-0.13	-0.11		-0.13
S + M + L	6.8%	-0.11	-0.04		+0.04	-0.03
S + M + I + L	4.4%	-0.19	-0.16	-0.18	-0.17	-0.18

Table 28: Calm-minus-panic z-score difference for the short leg. Nearly all entries are negative: in calm, the model shorts stocks deep in the left tail (strongly negative z-scores), while in panic it shifts toward shorting stocks closer to the cross-sectional average or even above it. The short leg's upward z-score shift in panic is the mirror image of the long leg's downward shift (Table 27), confirming that the regime signal reverses the selection direction on both sides of the portfolio.

Appendix H: Label Switching: Sign Correction Procedure

Because the four HMM features respond to stress in different directions, the expected direction of each feature during stress is determined empirically. For each feature j, the 95th percentile is computed separately on training months falling within known crisis windows (the dot-com bust of 2000–2002 and the Global Financial Crisis of 2007–2009) and on all remaining training months:

$$\text{sign}_j = \begin{cases} +1 & \text{if } q_{0.95}^{\text{crisis}}(z_j) \geq q_{0.95}^{\text{normal}}(z_j), \\ -1 & \text{otherwise.} \end{cases} \tag{40}$$

Features that spike upward during crises receive $\text{sign}_j = +1$; features that decline receive $\text{sign}_j = -1$. The panic state is then identified as the regime $k \in \{0, 1\}$ whose posterior mean vector $\hat{\boldsymbol{\mu}}_k$ best aligns with the crisis direction:

$$k^* = \arg\max_{k \in \{0,1\}} \sum_{j=1}^{D} \text{sign}_j \cdot \hat{\mu}_{k,j}. \tag{41}$$

The state labelled k^* is designated as "panic" and the other as "calm."

Appendix I: Convergence Diagnostics

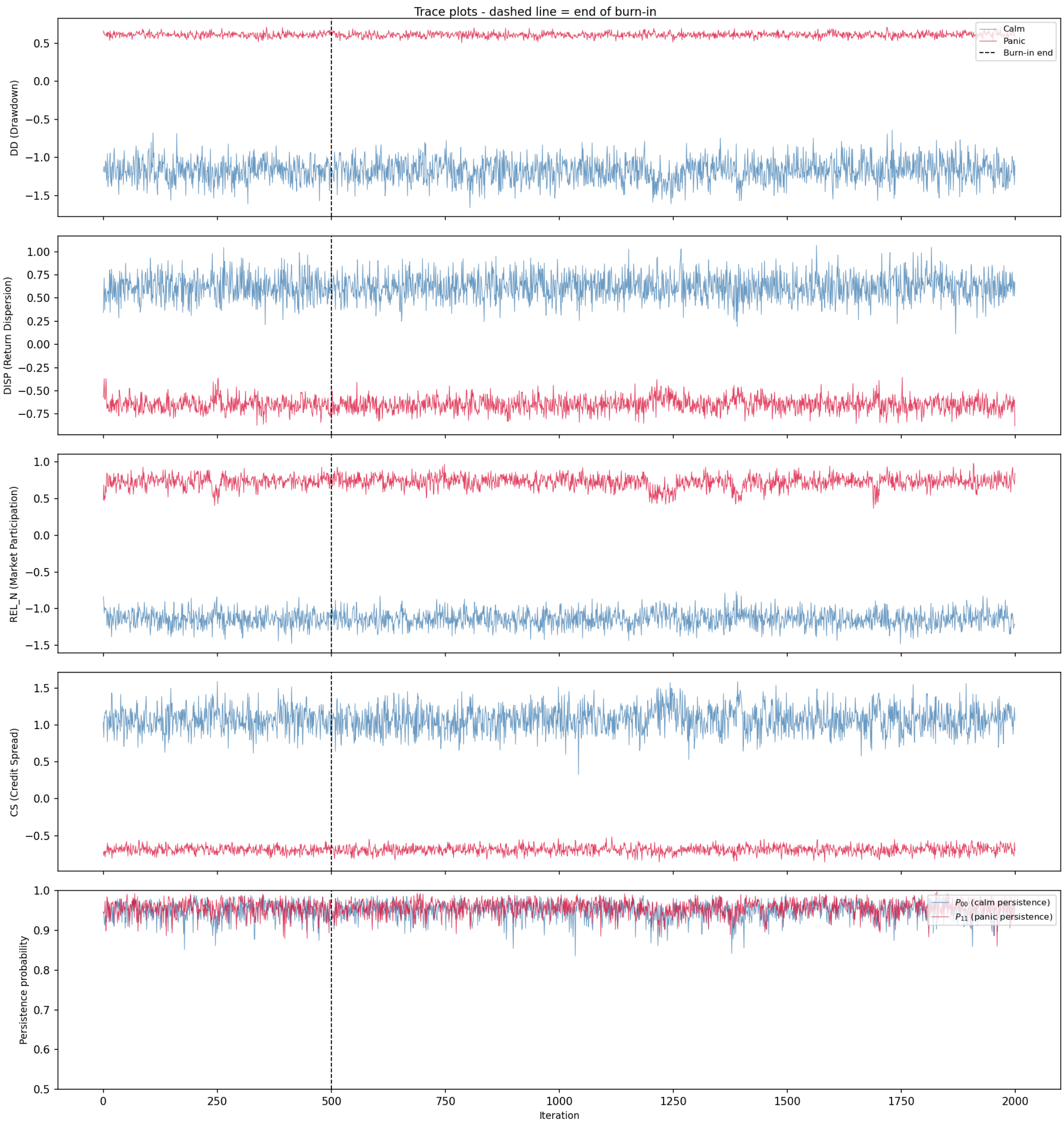

Figure 10: Trace plots of the Gibbs sampler. Each panel shows sampled regime means (calm in blue, panic in red) for one HMM feature. The dashed line marks the end of burn-in.

I.1 Gelman–Rubin Diagnostic

The Gelman–Rubin $\hat{R}$ statistic (Gelman et al., 2013) assesses convergence by comparing within-chain and between-chain variance across the five independent Gibbs sampler chains. For each parameter θ, $\hat{R} = \sqrt{\hat{V}/W}$, where W is the mean within-chain variance and $\hat{V} = \frac{n-1}{n}W + \frac{B}{n}$ combines within-chain variance with the between-chain variance B. Values near 1.0 indicate that the chains have converged to the same posterior distribution; $\hat{R} < 1.1$ is the standard threshold.

Parameter	$\hat{R}$	Status
μ_{calm}(DD)	1.000	OK
μ_{calm}(DISP)	1.000	OK
μ_{calm}(REL_N)	1.000	OK
μ_{calm}(CS)	1.000	OK
μ_{panic}(DD)	1.000	OK
μ_{panic}(DISP)	1.000	OK
μ_{panic}(REL_N)	1.000	OK
μ_{panic}(CS)	1.000	OK
P_{00} (calm persistence)	1.000	OK
P_{01} (calm $\rightarrow$ panic)	1.000	OK
P_{10} (panic $\rightarrow$ calm)	1.001	OK
P_{11} (panic persistence)	1.001	OK

Table 29: Gelman–Rubin $\hat{R}$ statistics for all HMM parameters, computed across 5 independent chains (1,500 post-burn-in draws each). All values are below 1.01, well within the convergence threshold of 1.1.

Appendix J: Fundamental Feature Definitions

Seven firm-level characteristics are computed from Compustat quarterly data (`fundq`), linked to CRSP via the CCM bridge table. All fundamentals are lagged four months from fiscal quarter end to ensure real-time availability.

- **Book-to-market (bm):** Book equity (`ceqq`) divided by market equity.
- **Return on equity (roe):** Income before extraordinary items (`ibq`) divided by book equity.
- **Gross profitability:** Revenue minus cost of goods sold, scaled by total assets: $(\texttt{revtq} - \texttt{cogsq})/\texttt{atq}$.
- **Asset growth:** Year-over-year growth in total assets: $\texttt{atq}_t/\texttt{atq}_{t-4} - 1$.
- **Leverage:** Total debt divided by total assets: $(\texttt{dlttq} + \texttt{dlcq})/\texttt{atq}$.
- **Earnings growth:** Year-over-year growth in earnings, transformed as $\text{sign}(g) \cdot \log(1 + |g|)$.
- **Log market equity:** Natural logarithm of market capitalisation (price $\times$ shares outstanding), lagged one month.

Fundamentals are winsorised at the 1st and 99th percentiles (5th and 95th for asset growth) using training-sample bounds.